Pricing Patterns on Dual-Mode E-Commerce Platforms*

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Abstract

Many of today's e-commerce platforms operate under a *dual* mode: they not only provide online marketplaces where third-party sellers compete and offer products, but also act as retailers selling products of their own. We study how dual-mode operation changes price competition on e-commerce platforms. Using an oligopoly pricing model, we show that, if the platform both earns commissions on third-party sales and retails on its own marketplace, price competition changes along two margins: platform retail presence can exert a more or less competitive effect on rivals' prices, and common cost shocks need not be passed through symmetrically across seller types. Using data from Amazon's U.S. marketplace, we exploit repeated within-product entry and exit episodes to show that third-party prices tend to respond modestly more to changes in other third-party sellers' availability than to changes in Amazon's own retail availability. Consistent with the model, we find asymmetric pass-through of commodity-cost shocks: third-party prices adjust more strongly than Amazon's retail prices, and cumulative pass-through for third-party sellers rises toward full transmission at longer horizons.

Keywords: E-Commerce, Dual Mode Platform, Online Pricing, Pass-Through

JEL Classification: D4, L1, E3

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1 Introduction

A growing number of e-commerce platforms – from Amazon and Walmart to Zalando and JD.com – operate in *hybrid* mode: they host third-party sellers while also selling directly to consumers themselves. Amazon is the leading example. Roughly 60% of sales on the platform are generated by third-party sellers (Federal Trade Commission, 2024), from which Amazon earns ad valorem commissions; yet Amazon also retails many of the same products on its own account. This dual role gives the platform a distinctive stake in downstream competition: it benefits when third-party sellers make sales, but it also competes with them for the transaction itself. As online retail expands[1] and platforms increasingly combine marketplace intermediation with their own retail activity, the competitive implications of hybrid operation have become increasingly important.

In this paper, we study how this distinctive market structure shapes price competition in product markets. Dual-mode platforms are not well described by the familiar case of a retailer selling private-label products alongside branded goods. In that setting, the retailer typically controls the retail prices of all products it carries. On a dual-mode platform, by contrast, third-party (3P) sellers set their own prices. Nor is a dual-mode platform well captured by models of common ownership: through percentage fees, the platform takes a share of sellers' revenues rather than their profits, and thus does not internalize their costs. These features create pricing incentives that are specific to dual-mode platforms and absent from more familiar market structures.

To study how price competition plays out on hybrid platforms, we combine a theoretical model with reduced-form evidence from Amazon. On the theory side, we develop an oligopoly pricing model to shed light on the core mechanism governing pricing decisions. A platform mediates price competition among differentiated-product retailers and, under a dual mode, also sells on its own account in direct competition with 3P sellers. We identify two platform-specific pricing wedges that arise when the platform sells its own products, relative to an otherwise comparable independent seller. First, because the platform earns an ad-valorem commission on 3P sales, raising its own price diverts demand toward 3P sellers and increases commission revenue. We refer to this as the revenue-sharing effect, which gives the platform an incentive to set a higher retail price. Second, because the platform does not pay the commission on its own sales, it faces a lower effective marginal cost than 3P sellers. We refer to this as the commission-saving effect, which gives the platform an incentive to set a lower retail price.

[1] See, e.g., https://www.census.gov/retail/ecommerce.html for the United States, https://ec.europa.eu/eurostat/statistics-explained/index.php?title=E-commerce_statistics_for_individuals for the European Union, https://www.ons.gov.uk/businessindustryandtrade/retailindustry/timeseries/ms6y/drsi for Great Britain, and https://dzswgf.mofcom.gov.cn/sjcx.html for China (accessed 28/08/2024).

These two distinctive pricing incentives render the platform's entry effect different from that of an independent seller, and in general ambiguous. In particular, when the revenue-sharing effect is strong, the platform's entry may exert little competitive pressure on 3P sellers. As a result, equilibrium prices may decrease only mildly (or not at all), contrary to the common belief that platforms like Amazon inevitably price-squeeze 3P sellers through their own offerings.

However, even if the platform's entry does not immediately price-squeeze 3P sellers, the commission-saving effect places 3P sellers at a structural competitive disadvantage and makes them particularly vulnerable to cost shocks. We find that, under an industry-wide common cost shock, the platform's price responds less than that of a 3P seller. This arises because the commission fee inflates 3P sellers' effective marginal costs, forcing them to adjust their prices more aggressively in response to cost shocks. Our theoretical pass-through results suggest that, in episodes of positive cost shocks, 3P sellers therefore face a trade-off: either absorb the higher costs and compress their margins, or raise prices more sharply than the platform, which causes them to lose traffic and market share. Since cost shocks occur repeatedly in practice, these temporary disadvantages can accumulate over time, enabling the platform to strengthen its position and expand its market presence from a dynamic perspective.

Empirically, our objective is to identify the pricing margins implied by the theory. To pair our model with reduced-form evidence, we collect a granular panel of prices for tens of thousands of products in narrowly defined product categories sold on `amazon.com`, the U.S.-facing storefront of Amazon.[2] We first document new descriptive facts on the degree of first-party (Amazon) selling within and across narrow product categories. We find that Amazon enters as a re-seller very selectively, is concentrated in high-turnover products, and, somewhat surprisingly, find that Amazon and 3P sellers enter and exit a given product repeatedly in manners that are not straightforward to predict with observables. Using hundreds of repeated within-ASIN episodes in which Amazon temporarily enters or exits as a retailer, we conduct short-run event studies of third-party sellers' pricing response upon entry (exit) by Amazon, and by 3P sellers, respectively. Under the assumption that the timing of these entry and exit episodes is conditionally exogenous – an assumption we can assess using pre-trends and auxiliary regressions – Amazon re-entry is followed by a 6% decrease in 3P prices, while Amazon exit is followed by a 3% increase. Comparing the pricing response of 3P sellers upon 3P entry (exit) to Amazon entry (exit), we find that the 3P seller prices move more strongly upon entry (exit) of other 3P sellers, as opposed to Amazon.

Our theory predicts that hybrid operation also changes pricing along a second margin,

[2] In the U.S., Amazon likely has a loyal customer base: 190 million customers are paid subscribers and receive membership benefits; see `https://cirpamazon.substack.com/p/updated-us-amazon-prime-member-estimate` (accessed 11/02/2026).

namely how prices react to common cost shocks. We estimate the impact of recent shocks to commodity costs, in particular, to cocoa and coffee beans, on consumer prices, using distributed lag regressions, hence without imposing structural assumptions. We find that Amazon passes on common cost shocks to a smaller degree than 3P sellers: pass-through to 3P seller prices is almost complete in long horizons, Amazon displays much lower pass-through. These empirical patterns hold robustly across different subsamples, and they line up with the commission-saving mechanism leading to lower cost pass-through for Amazon that our model puts forward.

Together, the entry/exit and pass-through patterns show that dual-mode operation affects both 3P price levels and the incidence of upstream cost shocks. Pass-through is informative about this incidence, as it measures the extent to which cost shocks are transmitted to consumer prices, rather than absorbed in seller margins.[3]

Related Literature

First, our paper contributes to the growing literature studying the implications of hybrid platforms for competition, innovation, and consumer welfare.[4] This literature has reached contrasting conclusions on whether allowing platforms to operate in dual mode harms consumers. Hagiu, Teh and Wright (2022) show that dual mode can benefit both the platform and consumers: the platform's own product "price-squeezes" a superior 3P seller, intensifying competition and thus increasing total transactions on the marketplace. In contrast, Anderson and Bedre Defolie (2024) find that the hybrid mode harms consumers because the platform optimally raises its commission fee relative to the pure marketplace mode in order to steer demand toward its own offerings. The higher fee reduces 3P participation and product variety, ultimately resulting in higher final prices.

Our paper differs from this literature in perspective and methodology. First, rather than endogenizing the platform's business-model choice, seller participation, or the commission fee, we take these elements as exogenous. This allows us to highlight the hybrid platform as a distinctive market structure and to examine how it shapes price competition. Second, by modeling marketplace competition as an oligopoly, we can capture rich strategic interactions between the platform's and 3P sellers' prices. This strategic interdependence – combined with the platform's

3For a recent popular-press discussion of cost shocks faced by Amazon sellers, see https://www.reuters.com/business/retail-consumer/some-amazon-sellers-are-pulling-out-prime-day-amid-trump-tariffs-2025-04-28/ (accessed 20/11/2025).

4This literature is partly motivated by the increasing prevalence of hybrid platforms across the economy as well as antitrust concerns associated with this business model; for example, the European Commission's cases on Amazon Marketplace and Amazon Buybox (cases AT.40462 and T.40703, see https://ec.europa.eu/competition/antitrust/cases1/202310/AT_40703_8990760_1533_5.pdf, or the U.S. FTC case, see https://www.ftc.gov/news-events/news/press-releases/2023/09/ftc-sues-amazon-illegally-maintaining-monopoly-power (both accessed 25/11/2025).

ability to share in its rivals' revenues through ad-valorem commissions – gives rise to the two key driving forces in our analysis: the revenue-sharing effect and the commission-saving effect. These incentives are absent in the earlier models: the homogeneous-preferences and fixed-fee framework of Hagiu et al. (2022) and the monopolistic-competition setting of Anderson and Bedre Defolie (2024) do not generate such pricing interactions. Finally, our paper offers a novel result by showing how the hybrid mode creates asymmetric pass-through of cost shocks between the platform and 3P sellers, and how this asymmetry can cumulatively disadvantage 3P sellers in dynamic competition. To the best of our knowledge, this channel has not been examined in the existing literature on hybrid platforms.[5]

Empirical evidence on pricing on Amazon remains relatively limited, especially on how hybrid operation shapes price competition and the transmission of cost shocks. Perhaps closest to our paper, a working paper by Crawford, Courthoud, Seibel and Zuzek (2022) uses proprietary data from Amazon Germany's Home & Kitchen department to study Amazon's first-party entry and shows that such entry is associated with lower 3P prices and modest market expansion. Other empirical work studies which products Amazon chooses to enter (Zhu and Liu, 2018), the existence and implications of self-preferencing and steering (Lee and Musolff, 2025; Chen and Tsai, 2024; Dendorfer and Seibel, 2026; Farronato, Fradkin and MacKay, 2023, 2025), the behavioral consequences of transparency about the Buy Box (Boldrini and Braulin, 2025), competition between offline stores and Amazon (He, Reimers and Shiller, 2022), and the consequences of regulating Amazon. In particular, Gutiérrez (2022) studies the welfare consequences of regulating Amazon, while Bennati (2026) and Bedre Defolie and Sokullu (2026) develop structural models of Amazon Marketplace competition to quantify the price, entry, and welfare effects of restricting Amazon's own retail activity. Relative to this literature, our paper studies two pricing margins of competition on Amazon: the short-run effect of Amazon's retail presence on 3P prices, and the pass-through of common cost shocks across seller types. We thereby naturally relate to a body of research that empirically estimates the pass-through of changes in exchange rates, commodity prices, taxes, tariffs, or other shocks in various real-world settings, oftentimes using reduced-form approaches. Most related may be Bettendorf and Verboven (2000); Nakamura and Zerom (2010) and Sangani (2026) who present pass-through estimates in the same product markets as us

[5]There exists further theoretical literature that deals with hybrid platforms, but does not address the pricing decisions that we study. For instance, this literature analyzes a hybrid platform's choice of entering into product markets (Etro, 2021); the optimal commission rate (e.g., Etro (2023)), or the optimal response of such a platform to collusive agreements among its sellers (Bisceglia and Padilla, 2023). Further theoretical work examines the incentives of platforms to "steer" consumers to their own products or to give biased recommendations (e.g., de Cornière and Taylor (2014); Drugov and Jeon (2019)), and estimates the existence of such steering empirically (see Lee and Musolff (2025) for Amazon, and Cure, Hunold, Kesler, Laitenberger and Larrieu (2022) in the context of a travel meta-search platform). Madsen and Vellodi (2023) study to what extent restrictions on a platform's use of data about the demand 3P sellers are facing, and the platform's subsequent entry decisions, can affect ex-ante product innovation incentives.

(coffee). Existing evidence of pass-through in e-commerce settings is macroeconomic, but offers opposite conclusions and does not model the competitive structure or pricing incentives of different seller types on e-commerce platforms (Cavallo, 2019; Gorodnichenko and Talavera, 2017; Goolsbee and Klenow, 2018).

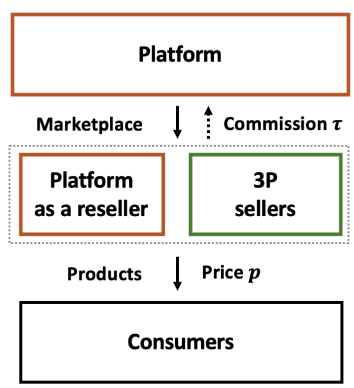
Outline. This paper is structured as follows. Section 2 describes the main institutional features of e-commerce platforms like Amazon, introduces the data we have collected, and provides stylized facts on seller participation and prices. Section 3 proposes a stylized model of price competition on a hybrid platform. Section 4 shows results on the prevailing price level and pass-through on Amazon. Section 5 concludes.

2 Institutional Setting and Stylized Facts

2.1 Hybrid Selling on Amazon Marketplace

Large e-commerce platforms increasingly combine a traditional retailing and a marketplace role in a hybrid structure: they operate a marketplace for 3P sellers and effectively tax these via a commission fee, while also selling first-party products in direct competition with those sellers (see Figure 1). This dual-mode structure has become prevalent across leading platforms and retailers worldwide, including Amazon, JD.com, and Walmart[6].

Figure 1: Illustration of a hybrid platform



Amazon – the world's largest e-commerce platform – has operated in dual mode since 2000. 3P sellers account for over 60% of sales on Amazon, and 560,000 sellers – predominantly small

[6] The hybrid model has emerged from both directions: pure marketplaces such as Taobao (see https://www.digitalwebsolutions.com/blog/alibaba-business-model/) have added first-party retail, while traditional retailers such as Walmart (see https://tinyurl.com/2pyvv3eh), Zalando (see https://www.brandongroup.it/en/zalando-partner-program-what-it-is-and-how-to-become-a-zalando-partner/) and Fnac (see https://www.lengow.com/integrations/fnac/; last accessed 01/03/2026) have opened their storefronts to 3P sellers.

and medium-sized firms – were active on the U.S. marketplace in 2021 (Federal Trade Commission, 2024).[7] Amazon levies an *ad valorem* commission (called "referral fee") on each mediated 3P transaction. The referral-fee schedule is public, varies by product category between about 5 and 20% of the final price (and, in some categories, may vary price tier), and changes infrequently.[8] Amazon's first-party retail activity includes some private-label offerings (e.g., Amazon Basics), but most first-party sales are in fact of branded products.

Unlike platforms such as eBay, Amazon aggregates all offers for a given product onto a single detail page, identified by an Amazon Standard Identification Number (ASIN). Sellers of the same ASIN compete primarily for the "Buy Box" – the prominent section on the top-right of the page – which captures the vast majority of sales and the allocation of which is governed by a proprietary algorithm (Chen, Mislove and Wilson, 2016; Lee and Musolff, 2025; Breslin, 2025). Offers from non-winning sellers are relegated to a less visible list ("Other Sellers on Amazon") below the Buy Box. Besides the ad-valorem referral fee, sellers can choose to outsource storage and shipping to Amazon ("Fulfilled by Amazon"), with an additional per-unit fulfillment fee[9], or avoid these fees by managing the logistics themselves ("Fulfilled by Merchant").

2.2 Data and Measurement

Data source. We collect data on prices of products sold on Amazon from Keepa (`keepa.com`), a company that provides data on prices, reviews, sales rank, and further both time-varying as well as cross-sectional attributes of products sold on Amazon by repeatedly scraping its product pages. Keepa updates its information multiple times per day, and its data have been used in previous academic literature (e.g., He et al. (2022); Gutiérrez (2022); Cabral and Xu (2021); He, Hollenbeck and Proserpio (2022); Bennati (2026)). Our sample period runs from January 2021 to March 2025, thereby excluding the period in which U.S. tariff changes may have influenced prices.

Sample. Our analysis is conducted at the level of a subcategory. We focus on categories that experienced sizable, well-identified upstream cost shocks in recent years and where demand fundamentals are relatively stable, in particular the ground coffee and cocoa categories (see Section 4).[10] We also collect data for other food (wheat pasta) and non-food subcategories (TVs, washers,

[7] Amazon sellers made an average yearly revenue of $250,000 in 2023; see `https://sell.amazon.com/blog/amazon-stats` (accessed 27/09/2024).

[8] See `https://sellercentral.amazon.com/help/hub/reference/external/GTG4BAWSY39Z98Z3` (accessed 27/11/2025); the most recent referral fee change on amazon.com we are aware of occurred in January 2024.

[9] FBA fee depends on product size and weight of the product, but not on its price. See `https://sellercentral.amazon.com/revcalpublic?ref_=sdus_pri_xscus_rc2&initialSessionID=135-2662106-4270236&ld=NSGoogle_SD_PRI_RC2&pageName=US%3ASD%3AFBA-main&ldStackingCodes=NSGoogle` (accessed 19/06/2026).

[10] Grocery products now account for a substantial share of Amazon's revenue, suggesting that these categories capture mainstream products. In 2024, Amazon CFO Brian Olsavsky noted "strong growth in items like every-

T-shirts, USB hubs, and SD cards). Within each product category, we treat each ASIN as the relevant product and restrict attention to the 10,000 products with the highest turnover, as proxied by both the current and the 180-day average sales rank[11]. Our sampling strategy lies hence between platform-wide studies that cover many departments with relatively few products per category (e.g., Chen and Tsai, 2024; Crawford et al., 2022; Lee and Musolff, 2025; Bedre Defolie and Sokullu, 2026) and single-market studies such as Bennati (2026).

In the Keepa data, whenever a product is being offered on Amazon, we observe the lowest 3P price fulfilled by Amazon (*FBA*), the lowest 3P price fulfilled by merchants (*FBM*), the lowest marketplace price overall (including Amazon retail), Amazon's own price, the total number of active sellers, the price displayed in the Buy Box, and the seller ID of the Buy Box winner. Additionally, we observe each product's sales rank. We use this to construct several variables that identify relatively high-turnover products: whether it is consistently highly-ranked in terms of sales rank (*popular)*; and a product's percentile in the sales-rank distribution. For food categories, we recover package sizes (weight in grams) from product titles using the OpenAI API.[12] We then construct a product-day panel using prices recorded at midnight, and aggregate these data to the product-week level by averaging all time-variant variables.

Table 1: Summary of sample, by category (ASIN-week panel; 222 weeks)

Category	Products (ASINs)	Avg. seller count (per ASIN-week)	ASIN-weeks with $\geq$ 2 sellers	With reviews (across ASINs)	Amazon private label (across ASINs)
Cocoa	1,432	2.3	32.4%	67.0%	0.2%
Coffee	14,532	2.5	35.8%	90.6%	0.3%
Pasta	7,709	2.5	42.0%	72.9%	0.4%
Washers	2,167	1.8	21.7%	22.8%	0.0%
TV	6,875	2.3	32.5%	64.8%	0.0%
Shirts	22,825	1.5	16.0%	92.4%	0.0%
USB hubs	7,269	2.2	20.0%	43.3%	0.4%
SD cards	12,851	1.3	14.3%	11.4%	0.0%

Notes: We track sampled ASINs daily from January 1, 2021 to March 31, 2025 and collapse the data to the ASIN-week level. We only include ASINs that have at least one week where Amazon or 3P has a valid price. The average seller count corresponds to the average number of sellers listing an ASIN in a week, computed over ASIN-weeks with at least one observed seller, and includes Amazon retail when present. "With reviews" and "Amazon private label" are product-level shares. The review indicator is measured from the latest observed product page and is therefore time-invariant within the sample.

Table 1 reports category-level summary statistics on sample coverage and seller participation.

day essentials. This includes items like health, beauty, and personal care, as well as nonperishable grocery," and stated that "the strength in everyday essential's revenue is a positive indicator that customers are turning to us for more of their daily needs." See https://www.fool.com/earnings/call-transcripts/2024/10/31/amazoncom-amzn-q3-2024-earnings-call-transcript/ (last accessed 01/02/2025).

[11]This "Best Sellers Rank" aggregates recent and historical sales relative to the category peer group. See https://sell.amazon.com/blog/amazon-best-sellers-rank for further information (accessed 02/09/2024).

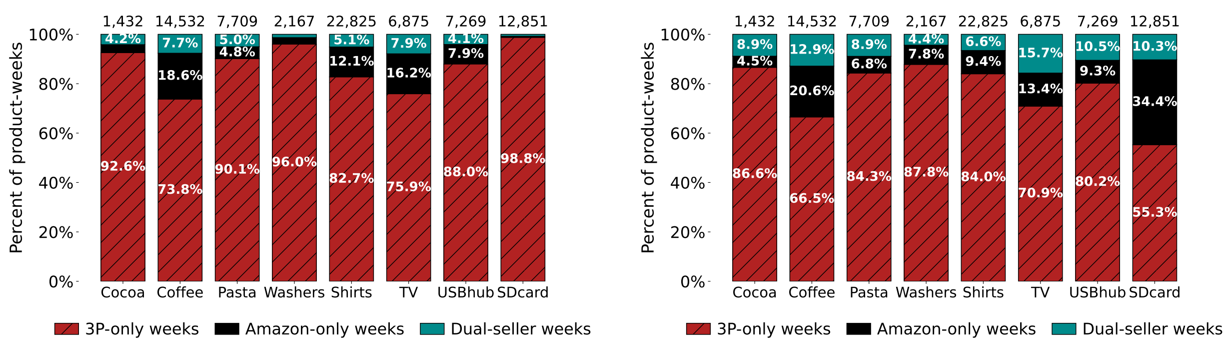
[12]Appendix A.2 provides details on how the data was built, and also shows that food products are sold predominantly in standard household sizes rather than bulk formats (Table A.2).

Product variety is substantial in every category: for example, the coffee category alone contains almost 40,000 distinct ASINs, 14,532 of which are in our dataset.[13] Competitive conditions also vary meaningfully across categories. The average number of sellers per ASIN-week ranges from 1.3 for SD cards to 2.5 for coffee, and the share of ASIN-weeks with at least two observed sellers ranges from 14.3% to 42.0%. Review prevalence also differs markedly across categories. By contrast, Amazon private-label products account for only a very small share of products in every category, never exceeding 0.4% of the products scraped.[14]

2.3 Stylized Facts

We document three stylized facts about Amazon's operation as a hybrid platform: the selectivity of Amazon's retail presence across categories and products, the transient nature of its presence over time, and the price levels of Amazon relative to 3P sellers.

Figure 2: Extent of hybrid selling across categories



(a) Equal-weighted product-weeks.

(b) Weighted by sales-rank percentile.

Notes: Each bar shows the share of ASIN-week observations that are 3P-only, Amazon-only, or dual-seller for a given category. Panel (a) weights each ASIN-week equally. Panel (b) weights ASIN-weeks by within-category sales-rank percentile, giving more weight to higher-turnover products. The sample covers ASINs observed weekly from January 2021 through March 2025. A product-week is classified as 3P-only (Amazon-only) if at least one 3P (Amazon) offer is recorded and the other side is absent; it is classified as dual-seller if both offer the products in the same week. The number of ASINs for each category is displayed above each bar.

Fact 1: Amazon retailing is selective and concentrated in high-demand ASINs. Figure 2 displays, for each category, the share of ASIN-week observations that are supplied exclusively by 3P sellers (3P-only), exclusively by Amazon (Amazon-only), or by both (dual-seller). Panel (a) weights each ASIN-week equally and shows that dual-seller states account for between about 0.8% and 7.9% of product-weeks across categories. 3P sellers are the exclusive suppliers in at least 73.8% of product-weeks in every category, while Amazon is the exclusive seller in between

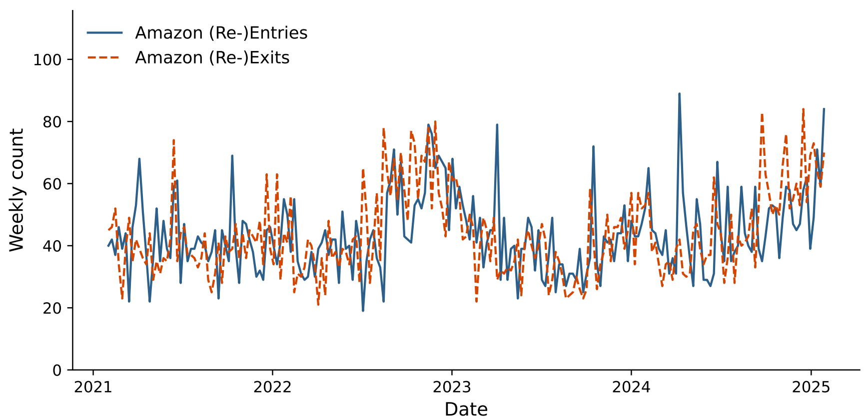
[13] Because we query the data repeatedly over time, the final number of products in a category can substantially exceed 10,000.

[14] Table A.3 contrasts these figures for the subsample of ASINs that ever experience first-party seller participation.

0.4% and 18.6% of product-weeks. Overall, Amazon's retail participation varies markedly across product categories: Amazon is active (either alone or alongside 3P sellers) in roughly 26% of ASIN-weeks in *Coffee* and about 24% in *TVs*, but in less than 2% of ASIN-weeks in *SD Cards*.

Panel (b) of Figure 2 weights ASIN-weeks by their within-category sales-rank percentile, giving more weight to higher-turnover products. Comparing the equal-weighted shares in panel (a) and the sales-rank-weighted shares in panel (b) shows that the prevalence of Amazon-only and dual-seller ASIN-weeks increases for all categories except for *Shirts*. For instance, in both *TVs* and *Coffee*, the share of ASIN-weeks in which Amazon is active (Amazon-only or dual-seller) rises from about 24-26% under equal weighting to around 30-34% under sales-rank percentile weighting. For *SD Cards*, Amazon-only and dual-seller states account for around 34% and 10% of sales-rank-weighted ASIN-weeks, respectively, which is markedly larger than before. Taken together, these patterns indicate that Amazon's direct retail presence is relatively more common among higher-turnover ASINs, while 3P sellers appear to account for most of the low-turnover "long tail" within categories[15].

Figure 3: Recurring Amazon entries and exits over time



Notes: The figure shows the weekly number of re-exits and re-entries by Amazon over time for the Coffee category. We exclude brands that account for more than 20% of aggregate Amazon entry or exit flows in the ten highest-activity weeks (one brand in the case of coffee), as those spikes would reflect systematic brand-level assortment changes rather than product-level entry or exit decisions (see Figure A.3). We only count a re-entry (re-exit) as such if they are followed by a spell of presence (absence) of 5 days or more (Figure is robust to other cutoffs).

Fact 2: Amazon retail participation changes repeatedly within the same ASIN. Amazon's retail participation varies along two margins. First, we observe a few changes in Amazon's

[15]Descriptively, we find a product's brand to be a strong predictor of whether Amazon is active as a reseller. See Appendix A.4 for these findings, as well as for a summary of what researchers have learned about the determinants of Amazon's presence as a retailer. These patterns should not be given a causal interpretation, as from the figures alone, it is unclear whether Amazon chooses to start selling popular ASINs, or whether Amazon's retail presence renders ASINs more popular. Nevertheless, these patterns are consistent with prior work documenting that Amazon tends to enter more popular product niches and that 3P sellers disproportionately populate the long tail (Hagiu and Wright, 2015; Zhu and Liu, 2018; Crawford et al., 2022; Bennati, 2026).

product assortment that are correlated at the brand level: instances in which Amazon enters or exits many ASINs of a given brand at the same time (see Figure A.3 for an example). These episodes are consistent with deliberate assortment decisions to start or stop reselling a brand. Second, we observe ASIN-level availability changes that, importantly, are *temporary* and tend to be dispersed over time. Figure 3 shows the weekly number of entries and exits by Amazon into ASINs at the weekly level over time for the Coffee category, showing that both (re-)entries and (re-)exits of Amazon are frequent and occur relatively evenly throughout the sample period. These entries and exits are in fact predominantly *repeated spells* Amazon's presence and absence as a reseller: the median ASIN has 3 Amazon-present spells and 4 Amazon-absent spells, with median spell durations of 3 and 9 days, respectively (see Appendix A.5). We observe similar patterns in other categories. These recurrent ASIN-level availability spells are consistent with lags in replenishment or stockouts, and will be used as a source of time series variation in the event studies of Section 4.1.

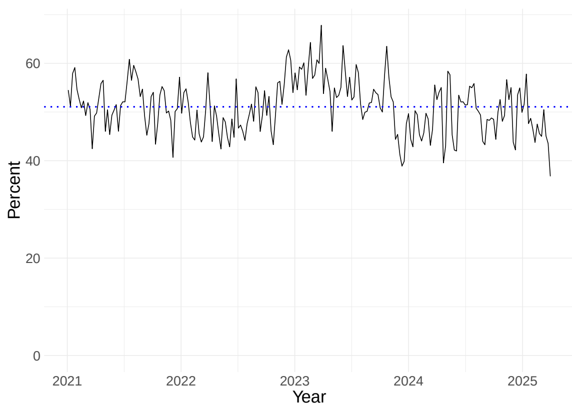
Fact 3: Conditional on same-ASIN overlap, Amazon prices are at or below comparable 3P prices. As Amazon sells a different set of products than 3P sellers, cross-ASIN price comparisons would confound pricing behavior with product selection. We therefore compare prices only within ASIN-weeks in which Amazon and at least one 3P seller simultaneously list the same product. In these dual-seller observations, Figure 4 shows that Amazon's price exceeds the lowest 3P offer in roughly one-third to one-half of ASIN-weeks across the displayed categories. The same pattern obtains in regressions with ASIN and year-week fixed effects (Tables A.9 to A.13): Amazon's price is generally statistically indistinguishable from, or below, the lowest 3P or FBA offer, with the clearest Amazon discount in coffee.[16] Thus, conditional on same-ASIN overlap, Amazon does not appear to occupy a systematically high-price position. We interpret this evidence descriptively: overlap itself is selected, and similar observed prices need not imply similar pricing incentives across seller types.[17]

Taken together, these facts show that Amazon's retail arm is a selective competitor: it is more prevalent on high-turnover ASINs, frequently alternates between presence and absence within the same ASIN, and, conditional on direct overlap, prices at or below comparable 3P offers. However, similar price levels need not imply similar pricing incentives. In the next section, we develop a theoretical framework characterizing the pricing incentives of Amazon and 3P sellers.

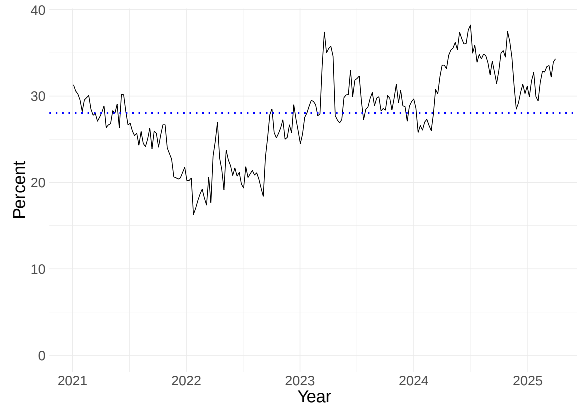
[16] Relative to FBM offers, Amazon is lower throughout, although this comparison is less direct because FBM prices include shipping costs.

[17] Appendix A.6 also reports weekly price-adjustment frequencies. Conditional on joint availability, differences across seller types are small; in coffee, Amazon makes somewhat more frequent but smaller price adjustments than 3P sellers.

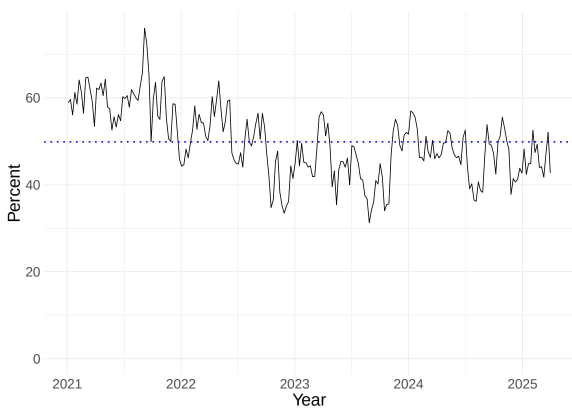
Figure 4: Percent of products offered by both 3P and Amazon for which Amazon's price is higher than the lowest 3P's price



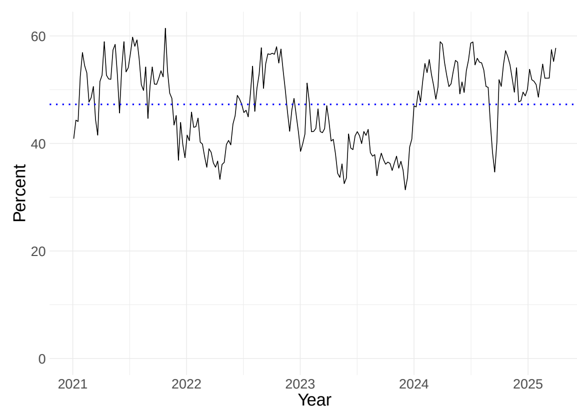
(a) Cocoa.



(b) Coffee.



(c) SD Cards.



(d) USB Hubs.

Notes: Plots use product-week observations in which both a 3P seller's, as well as Amazon's price is available (i.e., both seller types are active). Blue dotted line indicates median.

3 Theory

To interpret the forces driving pricing patterns and to decompose the mechanisms behind them, we develop a theoretical framework that models the platform's dual objective within a hybrid structure.

3.1 General Framework

Suppose that a monopoly e-commerce platform P operates a marketplace in a given product market. There are n retailers, each selling a differentiated product on the marketplace. The demand faced by retailer $i \in \{1, \ldots, n\}$, denoted $D_i(\mathbf{p}) = D_i(p_i, \mathbf{p}_{-i}) = D_i(p_1, \ldots, p_n)$, depends on its own price p_i and the prices of its rivals $\mathbf{p}_{-i}$. We assume a symmetric demand system. The marginal cost of retailer $i \in \{1, \ldots, n\}$ is c_i. The platform charges a commission fee τ on the revenue generated by each 3P seller on the platform. We take the commission fee as exogenous.

We consider two possible business models for the platform: the pure marketplace mode and the hybrid mode. In the pure marketplace mode, the platform hosts only 3P products. In the hybrid mode, the platform not only runs the marketplace but also sells its own product and competes directly with 3P sellers.

Timing: Under each business model of the platform, all retailers simultaneously choose their prices $p_i, i \in \{1, 2, \ldots, n\}$.

3.2 Pricing Incentives

In this subsection, we examine how the hybrid mode—that is, the platform's entry as a retailer—affects price competition in the product market, and in particular, how it influences the prices of 3P sellers relative to the pure marketplace mode. To isolate the pure effect of the platform's role as a retailer on pricing incentives from the entry effect, we keep the number of retailers fixed at n by assuming that retailer 1 represents the platform under the hybrid mode, and we denote its price by p_P.

In the pure marketplace mode, each retailer $i \in \{1, 2, \ldots, n\}$ chooses its price p_i to solve

$$\max_{p_i} \, [(1-\tau)p_i - c_i]D_i(\mathbf{p})$$
$$\Leftrightarrow \max_{p_i} \, (1-\tau)(p_i - \frac{c}{1-\tau})D_i(\mathbf{p})$$

The first-order condition is

$$D_i(\mathbf{p}) + (p_i - \frac{c_i}{1-\tau})\frac{\partial D_i(\mathbf{p})}{\partial p_i} = 0, \quad i = 1, \ldots, n \tag{1}$$

Therefore, under the pure marketplace mode, the commission fee effectively scales up each retailer's marginal cost by a factor of $\frac{1}{1-\tau}$. We denote the equilibrium under the pure marketplace mode by $\mathbf{p}^* = (p_1^*, \ldots, p_n^*)$, characterized by Equation (1).

In the hybrid platform mode, 3P retailers $i = 2, \ldots, n$ face the same profit maximization problem as under the pure marketplace mode. In contrast, the platform as retailer 1 solves the following problem

$$\max_{p_P} \, (p_P - c_P)D_P(\mathbf{p}) + \sum_{i=2}^{n} \tau p_i D_i(\mathbf{p}),$$

where its marginal cost is $c_P = c_1$.

The first-order condition is

$$D_P(\mathbf{p}) + (p_P - c_P)\frac{\partial D_P(\mathbf{p})}{\partial p_P} + \tau \sum_{i=2}^{n} p_i \frac{\partial D_i(\mathbf{p})}{\partial p_P} = 0 \tag{2}$$

We denote the equilibrium under the hybrid mode by $\hat{\mathbf{p}}^* = (\hat{p}_P^*, \ldots, \hat{p}_n^*)$.

Proposition 1. *When the platform operates in the hybrid mode, its pricing incentive as a retailer differs from that of an independent 3P seller in two respects:*

(1) ***Revenue-sharing effect****. Through the term $\tau \sum_{i=2}^{n} p_i \frac{\partial D_i(\mathbf{p})}{\partial p_P}$, the platform internalizes the positive impact of raising its own price on the sales of 3P sellers, and hence on its commission revenue. This gives it an incentive to set a higher price than p_1^*.*

(2) ***Commission-saving effect****. Because the platform does not pay a commission fee to itself, its effective marginal cost is lower than that of 3P sellers. This cost advantage provides an incentive to set a lower retail price than p_1^*.*

To facilitate comparative statics, we now impose the following specification.

Assumption 1. *All firms have identical marginal costs $c_i = c, \forall i \in \{1, \ldots, n\}$ and symmetric demand functions:*

$$D_i(\mathbf{p}) = \frac{1 - \gamma - (1 + (n-2)\gamma)\, p_i + \gamma \sum_{j \neq i} p_j}{(1-\gamma)\,(1 + (n-1)\gamma)},\ \forall i \neq j \in \{1 \ldots, n\}.$$

Under this assumption, we focus on the symmetric equilibrium under the pure marketplace mode. We denote the equilibrium price with n competing retailers by $p_M^*(n)$. Similarly, under the hybrid mode, we focus on the semi-symmetric equilibrium in which all 3P sellers charge the same price. We denote the platform's price by $\hat{p}_P^*(n)$ and the common price charged by 3P sellers by $\hat{p}_S^*(n)$.

Proposition 2. *Under Assumption 1, when $0 < \tau \leq \max\left\{0, 1 - \frac{2c(1+(n-2)\gamma)}{(n-1)\gamma}\right\}$, the revenue-sharing effect dominates the commission-saving effect. Consequently,*

(i) the platform charges a higher equilibrium price than 3P sellers: $\hat{p}_P^*(n) > \hat{p}_S^*(n)$;

(ii) the platform's entry exerts a smaller downward pressure on 3P sellers' prices than entry by an otherwise identical 3P seller: $\hat{p}_S^*(n) - p_M^*(n-1) > p_M^*(n) - p_M^*(n-1)$.

At first glance, this result may appear counterintuitive, as the revenue-sharing effect dominates when the commission rate τ is relatively low. In fact, this outcome is driven by the fact that the 3P sellers' effective marginal cost, $\frac{c}{1-\tau}$, increases nonlinearly with the commission rate. As a result, the commission-saving effect becomes increasingly important as τ rises. Only when the commission rate is sufficiently high does the commission-saving effect become strong enough to outweigh the revenue-sharing effect, inducing the platform to charge a lower price than 3P sellers.

Moreover, whenever the revenue-sharing effect dominates, the platform's entry exerts a smaller downward pressure on 3P sellers' prices than entry by an otherwise identical 3P seller. In other words, despite making the marketplace more competitive, the platform does not necessarily engage in aggressive price competition with 3P sellers by exploiting its cost advantage. This is precisely because the platform internalizes the positive impact of higher 3P sales on its commission revenue.

The intuition behind why the conditions for these two results coincide is straightforward. Under the linear demand specification, prices are strategic complements. Suppose the revenue-sharing effect dominates the commission-saving effect when evaluating the platform's marginal profit from increasing p_P at the pure-marketplace equilibrium $\mathbf{p}^*_M(n)$ (i.e., the left-hand side of Equation (2) is positive). In that case, the platform finds it profitable to raise its price above $p^*_M(n)$. Because prices are strategic complements, 3P sellers respond by raising their prices as well. Consequently, in equilibrium, we have $\hat{p}^*_P(n) > \hat{p}^*_S(n) > p^*_M(n)$. Conversely, if the commission-saving effect dominates, the platform finds it profitable to set a price below $p^*_M(n)$. 3P sellers then respond by lowering their prices. As a result, the hybrid-mode equilibrium features lower prices for all retailers than the pure marketplace equilibrium.

3.3 Asymmetric Pass-Through

In this subsection, we further investigate how the platform's special pricing incentives under the hybrid mode affect pass-through – that is, how the platform and 3P sellers respond differently to cost shocks.

To make the analysis concrete, we focus on the case with $n = 2$ and denote the two sellers by P and S. In addition, we assume the following linear demand function:

$$D_i(p_i, p_j) = \frac{1 - \gamma - p_i + \gamma p_j}{1 - \gamma^2}, i, j \in \{P, S\}; i \neq j,$$

where $0 < \gamma < 1$ measures the degree of product homogeneity.

Substituting the demand functions into the first-order conditions in Equation (1) and Equation

(2), the equilibrium prices under the hybrid mode are given by:

$$p_P = \frac{2c_P(1-\tau) + \gamma(1+\tau)c_S + (1-\gamma)(1-\tau)[2+\gamma(1+\tau)]}{(1-\tau)(4-\gamma^2(1+\tau))},$$

$$p_S = \frac{2c_S + \gamma c_P(1-\tau) + (1-\gamma)(1-\tau)(2+\gamma)}{(1-\tau)(4-\gamma^2(1+\tau))}.$$

Proposition 3. Pass-through matrix of cost shocks. *The pass-through from the platform's and the 3P seller's cost shocks to equilibrium prices is given by*

$$\rho = \begin{bmatrix} \frac{\partial p_P}{\partial c_P} & \frac{\partial p_P}{\partial c_S} \\ \frac{\partial p_S}{\partial c_P} & \frac{\partial p_S}{\partial c_S} \end{bmatrix} = \frac{1}{4-\gamma^2(1+\tau)} \begin{bmatrix} 2 & \frac{\gamma(1+\tau)}{1-\tau} \\ \gamma & \frac{2}{1-\tau} \end{bmatrix}.$$

(1) The platform has a lower own-cost pass-through rate than the 3P seller: $\frac{\partial p_P}{\partial c_P} < \frac{\partial p_S}{\partial c_S}$.

(2) The platform has a higher cross-cost pass-through than the 3P seller: $\frac{\partial p_S}{\partial c_P} < \frac{\partial p_P}{\partial c_S}$.

(3) When there is an industry-wide cost shock ε *to both sellers' costs, the platform experiences a smaller price increase than the 3P seller:* $\frac{dp_P}{d\varepsilon} < \frac{dp_S}{d\varepsilon}$.

Proposition 3 illustrates the crucial difference between own-cost and cross-cost pass-through under the hybrid mode. To understand the intuition, we rewrite the equilibrium conditions as

$$\begin{cases} p_P - (c_P + \tau p_S A_P) = -\frac{D_P}{\partial D_P/\partial p_P} & \text{(Platform)} \\ p_S - \frac{c_S}{1-\tau} = -\frac{D_S}{\partial D_S/\partial p_S} & \text{(3P Seller)} \end{cases}$$

Here, $A_P = -\frac{\partial D_S(\mathbf{p}^*)/\partial p_P}{\partial D_P(\mathbf{p}^*)/\partial p_P} = \gamma$ is the diversion ratio from the platform's product to the 3P seller, measuring the fraction of platform's lost sales that can be captured by the 3P seller when the platform increases its price. Hence, the $\tau p_S A_P$ can be interpreted as the opportunity cost faced by the platform when selling an additional unit of its own product[18].

From the equilibrium conditions, the difference in own-cost pass-through rates arises from the commission-saving effect. The commission fee scales up the 3P seller's effective marginal cost, but not the platform's. This amplification—captured by the term $\frac{1}{1-\tau}$—makes the 3P seller inherently more sensitive to its own cost shocks than the platform. In contrast, the platform exhibits a higher cross-cost pass-through due to the revenue-sharing mechanism. When a positive cost shock to the 3P seller increases its price p_S, the platform's opportunity cost $\tau p_S A_P$ of selling

[18]The need to write pass-through in the form of a matrix does not rely on a particular form of demand function, but only on the asymmetric market structure imposed by the hybrid mode. We have also derived a general form of pass-through matrix under a hybrid platform à la Weyl and Fabinger (2013), but omitted it due to it being difficult to interpret.

its product rises. This gives the platform an additional incentive to increase its own price. As the 3P seller earns only sales revenue, it does not internalize a similar channel when the platform's cost rises.

Since own-cost effects are first-order and dominate cross-cost effects, the platform's aggregate pass-through rate under an industry-wide cost shock is smaller than that of the 3P seller.

In short, Proposition 3 shows that the platform's price is more stable in the face of cost shocks, whereas the 3P seller's price is more volatile and vulnerable. This asymmetry in price stability has broader competitive implications. In episodes of positive cost shocks, 3P sellers face greater pressure to raise their prices, causing them to lose traffic and market share to the platform's product. Since cost shocks occur repeatedly in practice, these temporary disadvantages can accumulate over time, allowing the platform to strengthen its position and expand its market presence from a dynamic perspective.

4 Empirical Evidence

Our theoretical framework highlights a pricing asymmetry between the platform as a retailer and 3P sellers on a hybrid platform. Since the platform both earns commissions on 3P sales and does not pay commissions on its own sales, its pricing incentives differ from those of an otherwise similar 3P seller along two margins:

- **Prediction 1 (Effect on 3P Price Levels):** When the revenue-sharing effect dominates the commission-saving effect, platform retail entry exerts less downward pressure on incumbent 3P sellers' prices than entry by an otherwise identical 3P seller.
- **Prediction 2 (Asymmetric Pass-Through):** Due to the wedge created by commission fees, 3P sellers are more sensitive to cost shocks than the platform. Proposition 3 explicitly predicts that the platform's pass-through rate of common cost shocks should be strictly lower than that of 3P sellers ($\frac{dp_P}{d\varepsilon} < \frac{dp_S}{d\varepsilon}$).

In this section, we bring the model's predictions to the data. First, to study the price-level margin, we use the repeated within-ASIN changes in Amazon's and 3P sellers' retail availability and estimate event-time patterns in 3P prices around these changes. Second, to speak to the asymmetric pass-through prediction, we use shocks to commodity prices to compare the responsiveness of Amazon and 3P sellers to these exogenous input price changes. Although the result that the platform has a lower pass-through of common cost shocks than 3P sellers is derived in a linear duopoly set-up, we view it as capturing a general mechanism and use it to motivate our empirical analysis.

4.1 3P Pricing Reactions

4.1.1 Empirical Approach

The model predicts how Amazon's retail participation changes equilibrium prices. A cross-sectional comparison of ASINs with and without an observed Amazon retail offer would be difficult to interpret, because Amazon's retail presence is selected across products (see Fact 1 in Section 2.3) and likely correlated with product-specific demand, inventory, and 3P competitive conditions. Similar concerns apply for entry and exit of 3P sellers. We therefore use short-run within-ASIN variation in seller composition instead. For Amazon, we investigate the time around clean availability changes in which Amazon appears or disappears while 3P sellers remain active. Our comparison events are analogous changes in the number of active 3P sellers, while Amazon is absent.

Our empirical design relates to reduced-form studies of how entry affects prices (e.g., Arcidiacono, Ellickson, Mela and Singleton, 2020; Crawford et al., 2022; Fischer, Martin and Schmidt-Dengler, 2025), but the source of identifying variation differs from the canonical entry setting. The relevant changes on Amazon are frequent, short-lived, and reversible: a given seller type may appear or disappear on the same ASIN multiple times. We therefore employ data at the daily level and interpret "entry" and "exit" as changes in observed seller availability, not as permanent product-market entry or exit.[19] As treatment status is *reversible* rather than absorbing, the new difference-in-differences estimators designed for one-time staggered treatment adoption cannot be employed in this setting. Our empirical design instead estimates short-window event-time patterns around clean, isolated transitions in seller composition, as well as a more aggregated two-way fixed effect specification as secondary evidence.

A caveat of our analysis is that, in the data, we do not observe the full vector of 3P offers. We observe Amazon's posted price when Amazon is active, and for 3P sellers we observe the lowest FBA and FBM offers. We therefore interpret the empirical outcome in this subsection as the 3P price frontier under either FBA or FBM. Under the additional assumption that the frontier price is a sufficient empirical proxy for the relevant competitive 3P price, movements in this frontier provide reduced-form evidence on the price-level margin emphasized by the model.

Sample. We implement the main analysis in the coffee category, where Amazon retail participation is frequent and where most products have reviews (see Table 1).[20] For all analyses in this subsection, we restrict attention to ASINs that exhibit meaningful variation in seller compo-

[19] We hence use "entry" and "exit" as shorthand for changes in observed seller availability within an ASIN, rather than permanent product-market entry or exit.

[20] As opposed to apparel or electronics, this category is also not exposed to contemporaneous entry by low-cost import platforms such as Temu or Shein.

sition: specifically, we require that, at any point in our sample, (i) Amazon and 3P sellers are simultaneously active for at least 10 consecutive days, and (ii) only 3P sellers are active for at least 10 consecutive days. We moreover exclude the brand that displays a brand-level assortment change in our period of observations (as discussed Section 2.3).

Event definition for event study approach. As "spells" of seller presence can be short and are not always preceded or followed by a stable period of 3P-only selling, we next restrict attention to "clean" events with well-defined pre- and post-event windows. We define a clean "Amazon entry" event as a sequence in which (i) a 3P price is observed for at least K consecutive days prior to entry, and (ii) both an Amazon price and a 3P price are observed for the subsequent K consecutive days. We define "Amazon exit" analogously. In the same vein, we define an "3P entry" event as a sequence in which only 3P sellers sell around a period of K consecutive days prior and after a 3P seller entry event, and likewise for 3P exit. In the baseline specification, $K = 10$. The Appendix reports analogous results for alternative horizons.

Econometric specification. We perform two types of sharp event studies (Kleven, Landais and Søgaard, 2019) around symmetric windows of these events. For each event e, let $i(e)$ denote the ASIN associated with the event, and let T_e denote the event date. Define relative event time as

$$r = t - T_e$$

Our first specification analyzes Amazon entry and exit only, to understand whether Amazon's presence and absence already is followed by any pricing reaction in the lowest 3P seller's price. We write:

$$y_{er}^{m} = \sum_{j=-10,\, j\neq -1}^{10} \beta_j^m \mathbf{1}\{r = j\} + \alpha_e + \varepsilon_{er}^{m} \tag{3}$$

where α_e is an event fixed effect[21], and y_{er}^{m} the outcome variable, in particular the logarithm of the lowest 3P seller's price, $p_{it}^{3P,m}$, for $m \in \{\text{FBA}, \text{FBM}\}$. The coefficient β_j^m measures the change in the log lowest 3P price at event day j, relative to day -1, within the same event.

Second, we conduct a stacked event study that contains both 3P sellers' and Amazon's entry (exit), respectively, in the same regression, to be able to cleanly compare Amazon's impact with that of 3P sellers. Let

$$A_e = \begin{cases} 1 & \text{if event } e \text{ is an Amazon event,} \\ 0 & \text{if event } e \text{ is a 3P seller-count event.} \end{cases}$$

[21] This fixed effect is more granular than ASIN fixed effects would be, given that ASINs may undergo relevant entry or exit events multiple times. We also conducted event studies using calendar-date and ASIN fixed effects, with qualitatively very similar results.

We estimate, separately for entry and exit:

$$y_{er}^m = \sum_{\substack{j=-10 \\ j \neq -1}}^{10} \left[\beta_{j,A}^m \mathbf{1}\{r = j\} \mathbf{1}\{A_e = 1\} + \beta_{j,3P}^m \mathbf{1}\{r = j\} \mathbf{1}\{A_e = 0\} \right] + \alpha_e + \varepsilon_{er}^m \tag{4}$$

For the stacked entry regression, $A_e = 1$ denotes Amazon entry and $A_e = 0$ denotes 3P entry; for the stacked exit regression, $A_e = 1$ denotes Amazon exit and $A_e = 0$ denotes 3P exit. $\beta_{j,A}^m$ is the Amazon-event path at event time j, and $\beta_{j,3P}^m$ is the 3P-event path at event time j. The formal Amazon-vs.-3P difference at event time j is $\beta_{j,A}^m - \beta_{j,3P}^m$.[22]

A causal interpretation requires as underlying assumptions that (1) the date of entry/exit is exogenous, and (2) that prices would evolve smoothly absent Amazon's entry or exit. We can visually assess the validity of these assumptions using the event study estimates.

Predictors of the timing of Amazon entry and exit. Two features of the data are useful for assessing the plausibility of our design. Our Fact 2 above demonstrates the substantial within-ASIN variation in Amazon retail availability: for ASINs that switch Amazon-presence status, Amazon-present and Amazon-absent spells recur frequently and vary considerably in duration (see Appendix A.5). Second, Figure 3 shows that entry and exit events do not appear to be concentrated on a small number of calendar dates. This suggests that these events are not driven by platform-wide shocks.

To assess whether Amazon's event-study-qualifying availability changes are systematically anticipated by observables, we estimate hazard (logit) models for the probability that Amazon enters or exits an ASIN on the following day, conditional on the product's current seller regime. Reassuringly, the estimates in Tables A.7 and A.8 suggest that lagged price levels, recent price changes, or changes in sales rank have little predictive power for "qualifying" Amazon entry and exit. In contrast, Amazon entry and exit is more likely on certain weekdays, and more likely for ASIN-days with greater marketplace activity. We therefore interpret the hazard models as evidence against the concern that our event-study estimates simply reflect pre-existing price dynamics. Instead, it is possible that Amazon's availability changes are related to demand, inventory, and replenishment changes.

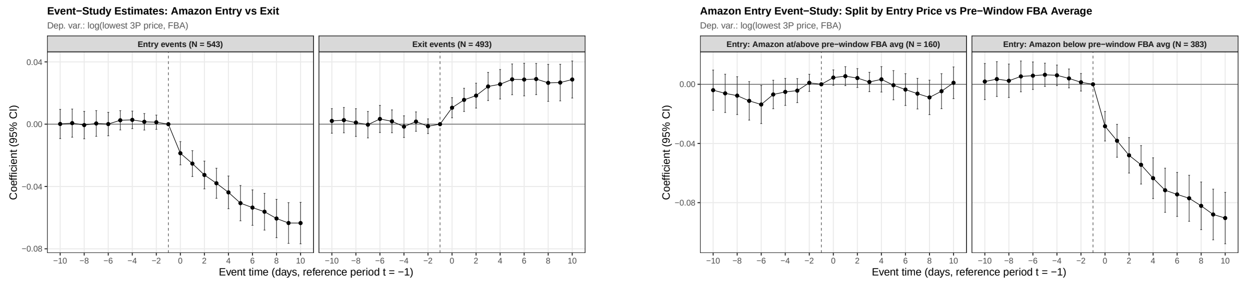
4.1.2 Results

Amazon entry and exit.

We first estimate Amazon-availability event studies (equation 3), using the lowest 3P seller

[22] To make 3P seller and Amazon events comparable, we additionally demand that the count in the total number of sellers does not change other than in $t = -1$.

Figure 5: Event study of entry and exit by Amazon as a seller into a given ASIN. Coffee data (specifications (1) and (3) of Table A.16).



(a) Lowest 3P price under FBA over event window.

(b) Split by Amazon's price level at entry.

Notes: Figures show coefficient estimates stemming from Equation 3, as well as 95% confidence intervals computed using two-way clustered standard errors (ASIN and calendar-date). Panel (a) displays specifications (1) and (3) of Table A.16. Panel (b) shows a sample split of specification (1) using, first, observations in which Amazon enters with a price that is at least as high as the pre-event average lowest FBA price, and second using observations in which Amazon enters with a price that is lower. All regressions are estimated using "clean" events in the daily coffee data, as defined earlier.

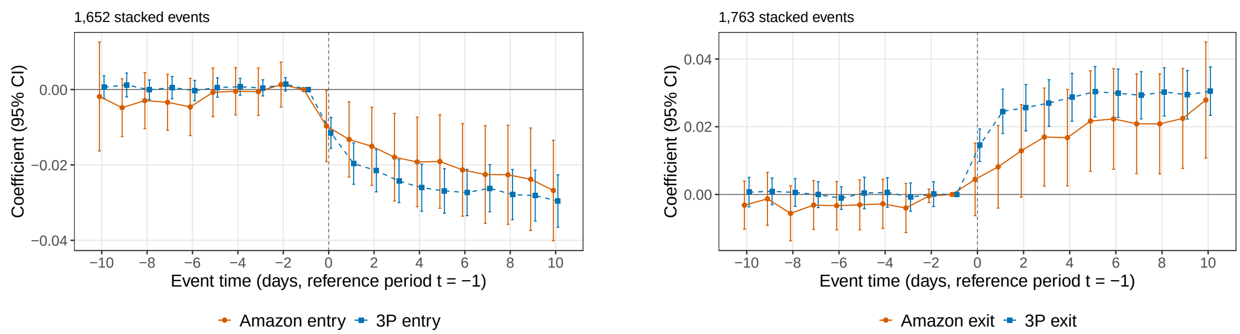
price under FBA as an outcome variable. The result is shown in Figure 5 (a); coefficients and fit statistics for both FBA and FBM prices are displayed in Table A.16. For entry events (columns (1) and (2), we observe pre-event coefficients that are jointly insignificant, and a strong decline in the 3P seller's price by almost 6% in the case of the FBA price (and by 3% in the case of the FBM price; see Figure A.6). Looking at exit events, we find the opposite pattern: mostly flat pre-trends are followed by an increase in the price by up to 3% in the case of FBA prices (similar in the case of FBM prices). Focusing separately on events at which Amazon enters with a relatively high (low) price yields striking and intuitive results (panel (b)): price declines upon entry are significant only if Amazon enters with a price that is lower than the average FBA price in the pre-window period; the analogous finding arises in the case of exit events.

In the Appendix, Figure A.7 shows that during the pre-period, sales rank stays roughly flat, while it decreases upon Amazon entry (an indicator for an increase in demand), and increases upon Amazon exit. Figure A.6 shows that the pricing reaction is possibly slightly stronger in more competitive environments, and that the seller count does not vary much around the Amazon entry or exit events. To summarize, Amazon entry and exit triggers reasonably large pricing reactions in 3P sellers' lowest price, with reactions being higher under FBA, which is intuitive as this is the closer substitute.

Contrasting Amazon with 3P seller entry and exit. Our key objective is to compare 3P sellers' pricing response to Amazon's entry compare to the response to 3P sellers' entry, which we can do in a clean way by estimating equation 4. The baseline results are shown in Figure 6 (we focus on pricing response of sellers operating under FBM due to the absence of pre-trends).

It becomes clear that, while Amazon entry appears to lower the 3P price frontier, it does not do so for more than 3P entry. 3P exit raises the price frontier strongly, while Amazon exit is weaker. Figure A.8 shows the analogous figure for sellers operating under FBA. Table 2 summarizes the findings for all event types and all price outcome variables. The direct contrast between post-period coefficients for Amazon and 3P entry show that Amazon entry incurs a significantly lower reaction in 3P prices under FBA, although this difference is not significant for the case of FBM. We also obtain very similar results when extending the event windows to $K = 15$ (Figure A.9).

Figure 6: "Stacked" event study of entry and exit by Amazon and 3P seller; outcome variable: $p^{3p,FBM}$



(a) Entry. (b) Exit.

Notes: Figures show coefficient estimates stemming from Equation 4 with the lowest 3P sellers' price under FBM as an outcome variable, as well as 95% confidence intervals computed using two-way clustered standard errors (ASIN and calendar-date). Panel (a) uses entry events; Panel (b) uses exit events. All regressions are estimated using "clean" events in the daily coffee data, as defined earlier.

Table 2: Average post-event effects in stacked Amazon-vs-3P event studies

Event	Outcome	Window	3P event	Amazon event	|Amazon| − |3P|	p-value: |Amazon| = |3P|	Events: Amazon / 3P
Entry	FBA	0–10	-3.11*** (0.35)	-1.63*** (0.56)	-1.48** (0.65)	0.023	147 / 943
Entry	FBA	3–10	-3.45*** (0.38)	-1.89*** (0.61)	-1.56** (0.71)	0.028	147 / 943
Entry	FBM	0–10	-2.45*** (0.29)	-1.92*** (0.57)	-0.52 (0.61)	0.387	207 / 1445
Entry	FBM	3–10	-2.70*** (0.31)	-2.17*** (0.62)	-0.54 (0.67)	0.420	207 / 1445
Exit	FBA	0–10	2.89*** (0.38)	0.26 (0.26)	-2.63*** (0.46)	0.000	222 / 1001
Exit	FBA	3–10	3.02*** (0.41)	0.25 (0.32)	-2.77*** (0.51)	0.000	222 / 1001
Exit	FBM	0–10	2.73*** (0.33)	1.78*** (0.68)	-0.95 (0.76)	0.209	232 / 1531
Exit	FBM	3–10	2.94*** (0.35)	2.12*** (0.74)	-0.82 (0.82)	0.317	232 / 1531

Notes: Estimates are average event-time coefficients over the indicated post-event window, relative to $t = -1$. Estimates and standard errors are reported as log points multiplied by 100. 3P denotes the third-party seller-count event path; Amazon denotes the implied Amazon-event path. |Amazon| − |3P| compares the absolute magnitudes of the implied Amazon-event and third-party seller-count event paths; negative values indicate that the Amazon-event response is smaller in absolute value. Standard errors are computed from the full two-way clustered covariance matrix, clustered by ASIN and calendar date. Event counts are unique event IDs in the complete-outcome estimation sample.

This clean event-study comparison is implemented in coffee, where the density of events is most favorable. However, a simpler within-ASIN two-way fixed effects specification across the broader scraped categories yields uniformly negative associations between Amazon presence and 3P prices (Table A.17). Across all product categories, the presence of Amazon as a competitor within the same ASIN appears to be associated with statistically significant lower FBA prices, with magnitudes ranging from 2.9% for TVs to 9.8% for pasta. Intuitively, when regressing on the lowest FBM prices, the magnitude of all effects becomes smaller, with the exception of TVs (Appendix Table A.17): this is aligned with the idea that sellers operating under FBA may be more direct competitors with Amazon than sellers operating under FBM.

4.1.3 Discussion

A limitation of our data is that, as mentioned, we do not observe the type of 3P seller that enters or exits.[23] The estimates should hence not be interpreted as identifying all equilibrium prices or as a test of the model's comparative statics, although they show that Amazon's retail availability affects the competitive price frontier faced by consumers and 3P sellers.

By focusing our analysis on products which Amazon eventually enters in, we avoid having to face a possible selection effect that stems from the fact that Amazon enters products selectively (as documented in Fact 1 above). Moreover, the absence of economically meaningful pre-trends in prices and sales rank across most specifications, and the weak predictive power of observable pre-event dynamics in the hazard models, make it unlikely that the estimated price responses merely reflect anticipatory repricing or predictable changes in demand. We therefore view the estimates as suggestive of short-run causal responses, conditional on our identifying assumptions.

4.2 Asymmetric Pass-Through of Cost Shocks

4.2.1 Empirical Approach

We next study the pass-through margin of competition on a hybrid platform: whether Amazon and 3P sellers differ in their transmission of common cost shocks. Our empirical strategy exploits plausibly exogenous variation in cocoa-bean and coffee-bean prices, and traces these shocks into their most direct downstream retail counterparts on Amazon, namely baking cocoa and ground coffee.[24] Accordingly, our pass-through estimates are reduced-form: they trace equilibrium retail price responses to plausibly exogenous commodity-cost shocks without imposing a full model of demand, seller conduct, or platform behavior. Prior work has also used coffee-bean price shocks

[23] In principle, this would be available via *Keepa*, but not over the time period studied in this paper.

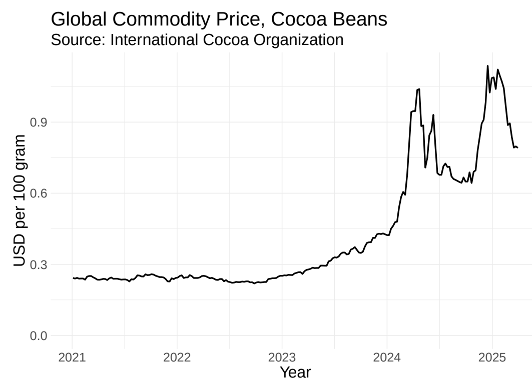
[24] We do not study other products affected by cocoa-bean prices, such as chocolate bars, because these products often involve a large share of so-called "local costs" (Nakamura and Zerom, 2010), such as advertising expenditures. In addition, chocolate manufacturers have reportedly adjusted product recipes in response to input-price shocks (see `https://www.ft.com/content/169c9fca-1a7a-4c0f-84e9-398777335629`, last accessed 26/02/2025), which further limits the suitability of these products for studying price pass-through.

to study pass-through; see Bettendorf and Verboven (2000), Nakamura and Zerom (2010), and Sangani (2026).

We obtain commodity price data at the daily level from Trading Economics (`tradingeconomics.com`), and (for cocoa) from the International Cocoa Organization (`www.icco.org`; see Appendix A.1). Figure 7 plots these corresponding prices. Reportedly, the observed hikes in price during our period of observations were due to bad weather and speculation, as opposed to aggregate demand shocks.[25]

Figure 7: Commodity prices: coffee beans and cocoa beans, January 2021-March 2025

(a) Coffee.



(b) Cocoa.

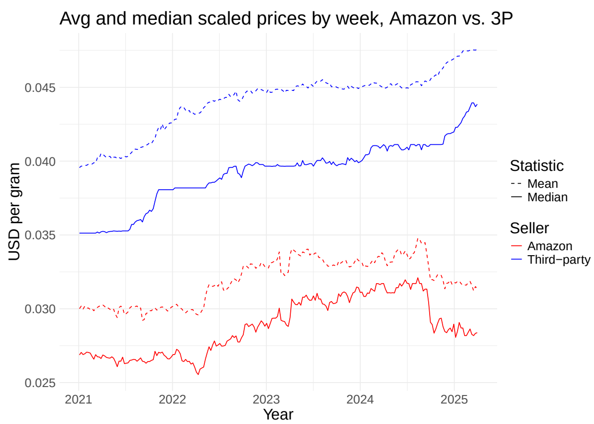
Descriptive analysis. Panels (a) and (b) of Figure 8 present the Year-week-sellertype fixed effects estimated from regressions of prices on year-week-sellertype and manufacturer-brand fixed effects, where sellertype can be either Amazon, or 3P sellers. For cocoa, there is no statistically significant difference between Amazon's prices and those of 3P sellers in any week. For coffee, Amazon's prices are significantly lower than 3P sellers' prices, even after controlling for manufacturer-brand fixed effects. However, this discrepancy is likely driven by differences in product selection: while some 3P sellers seem to specialize in high-end brands or premium roasts, Amazon appears to focus more on standard, high-turnover brands, based on anecdotal evidence.

To better illustrate price changes of particular products, in panels (c) and (d) we plot the corresponding seller-specific chained Fisher price indices. The underlying idea of the Fisher price index, frequently used in macroeconomics, is to compare prices only for ASINs that exist in two adjacent weeks, build a weekly link (so that new or retired items do not distort the level), and chain the links across the whole period. In both categories we observe a clear upward drift in the overall price level, consistent with the large increases in input prices documented in Figure

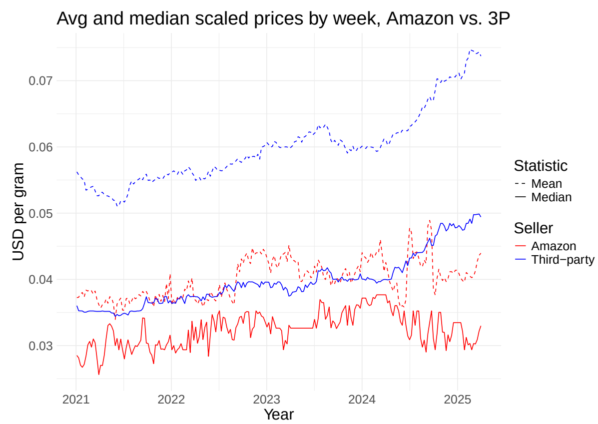
[25]See `https://www.ft.com/content/4ea3116d-6a55-4d14-8a47-8988209dc1e5` and `https://www.ft.com/content/8a917bea-8ce3-11e3-ad57-00144feab7de`, both accessed 10/10/2025.

7. The Amazon and 3P indices tend to co-move over time, indicating that, once we hold the set of products within each seller type approximately constant, 3P sellers might be raising prices somewhat more than Amazon. Since both indices are normalized to 100 in the first week of 2021, deviations between the two series at later dates can be interpreted as cumulative differences in price changes rather than in initial price levels.[26]

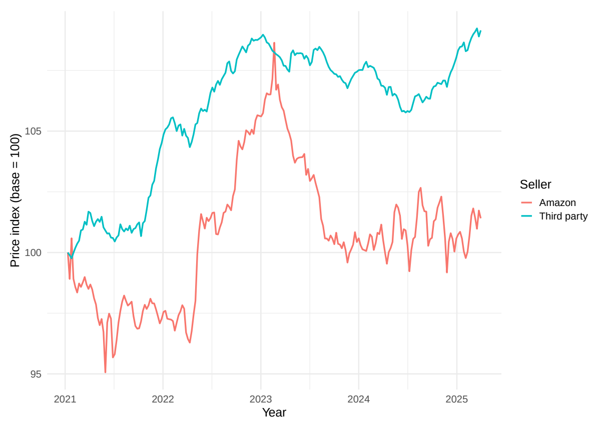
Figure 8: Prices over time: descriptives.



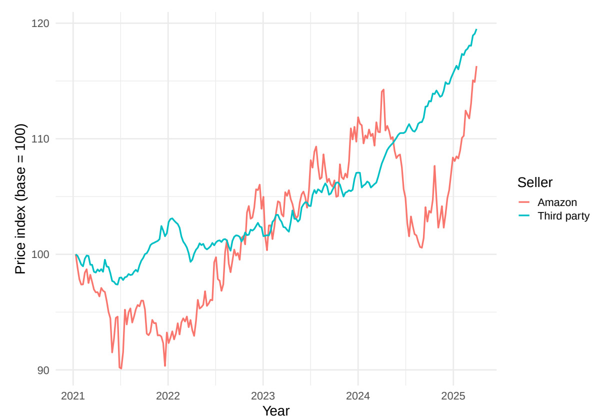
(a) Coffee, average and median prices per gram.



(b) Cocoa, average and median prices per gram.



(c) Coffee, chained Fisher price index.



(d) Cocoa, chained Fisher price index.

Notes: Plots (a) and (b) display average and median prices for Amazon and 3P sellers, respectively. In panels (c) and (d) we plot seller–specific, chained Fisher price indices. For each seller type and pair of adjacent weeks t and $t+1$, we first restrict attention to the set of ASINs that are sold in both weeks and compute a two-period "Fisher link", defined as the geometric mean of a Laspeyres-type and a Paasche-type price relative across these overlapping products. We give equal weight to each overlapping ASIN. We then chain these weekly Fisher links over time and normalize the index to 100 in the first week of 2021, so that the resulting series traces the cumulative evolution of prices for a constant set of products within each seller type, abstracting from entry and exit in the assortment.

Econometric specification. We estimate pass-through using a standard distributed-lag speci-

[26] A common concern is that rising prices can affect package sizes ("shrinkflation"); however, we do not find any evidence of a change in the prevailing package size distribution over the sample period.

fication, which is widely used in the pass-through literature (e.g., Sangani (2026); Nakamura and Zerom (2010); Conlon and Rao (2020); Alvarez, Cavallo, MacKay and Mengano (2024); Besanko, Dubé and Gupta (2005)). This specification provides a flexible reduced-form characterization of the dynamic response of retail prices to commodity-cost shocks.

For product i observed in week t, our baseline specification is:

$$\Delta p_{it} = \sum_{k=0}^{K} \beta_k \Delta c_{t-k} + \gamma_i + \epsilon_{it} \tag{5}$$

where Δp_{it} is the change in product i's price from week t-1 to week t, measured in dollars per gram, and Δc_{t-k} is the change in the relevant commodity price, also measured in dollars per gram, from week $t-k$ to week $t-k-1$. Including K lags (for example, $K = 6$ or $K = 12$) allows us to capture delayed pass-through of upstream cost shocks. Because the dependent variable is first-differenced, time-invariant product characteristics, whether observed or unobserved, are differenced out by construction. The product fixed effects γ_i allow each ASIN to have its own average drift in price changes, which accommodates persistent differences in repricing behavior across products.

Given that both downstream prices and upstream commodity costs are measured in the same units in our data, the levels specification has a direct economic interpretation: $\sum_{k=0}^{K} \beta_k$ measures the cumulative pass-through of a change in commodity cost per gram into product prices per gram. This is also the specification we emphasize in light of recent evidence in Sangani (2026), which shows that for common commodity shocks, pass-through may appear incomplete in logs even when it is close to complete in levels. We therefore use the levels specification as our baseline and report the analogous log specification in Appendix A.8.

We estimate Equation 5 using ordinary least squares. In our baseline regressions, we use either Amazon's price, or the overall lowest 3P seller's price (as defined in Appendix A.2), as the dependent variable. We confirm that first differences in commodity prices are stationary for both cocoa and coffee using an augmented Dickey-Fuller test. We also confirm that the distribution of package sizes of sold products does not change significantly over time, mitigating concerns of "shrinkflation" as a response to cost shocks. We estimate the regression separately for each product category and for prices set by Amazon itself, and for prices set by 3P sellers.

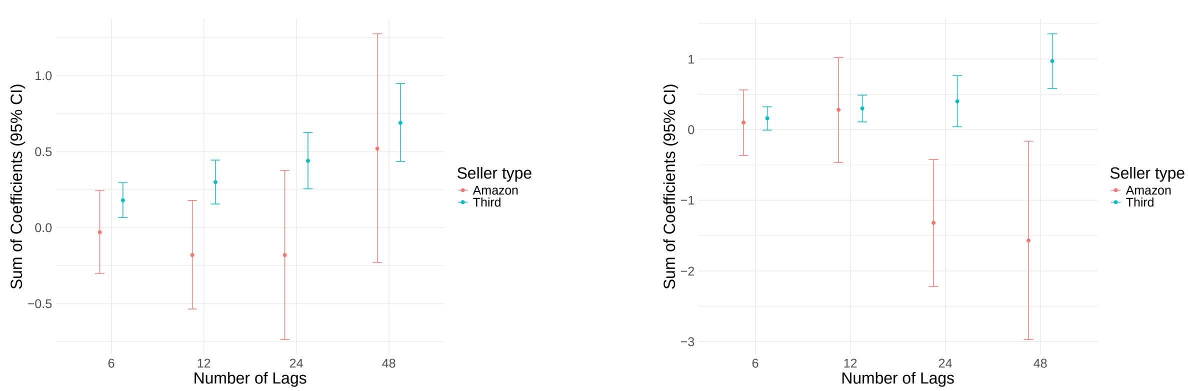
4.2.2 Results

Estimates for shorter and longer-term pass-through, $\sum_{k=0}^{K} \beta_k$, of cocoa and coffee in level and with K varying between six and 48 weeks are visualized in Figure 9 and documented in Table 3. We first consider pass-through on 3P sellers' prices. We find positive and significant coefficients that are in line with prior work in economic magnitude (Nakamura and Zerom, 2010; Sangani, 2026). Our baseline coefficients monotonically increase in magnitude as we include more lags, and approach 1 especially for cocoa when moving to a lag of 48 weeks. We find notably different results when considering only Amazon's prices: for the case of cocoa, we obtain significantly negative pass-through estimates after 24 or 48 weeks. For the case of coffee, our estimates tend to be insignificant.

Importantly, these results hold when estimating the same regression using a "stacked" regression on the subset of ASIN-weeks sold by both seller types, using a seller-type dummy that is interacted with the main regressor. While this "common" sample is much smaller, the results for coffee confirm that Amazon's cumulative pass-through is significantly lower than that of third-party sellers at medium and long horizons. For cocoa, the common-sample estimates are noisier, and the Amazon–3P difference is statistically significant only at the 48-week horizon.

These findings qualitatively also hold across different specifications and different sub-samples. For instance, they hold in the subset of products which have at least 36 weeks of consecutive 3P (or Amazon, respectively) retail presence (Table A.25); moreover, pass-through tends to be higher for sellers operating under FBA, as opposed to sellers operating under FBM (Table A.21). The main noteworthy exception arises when conditioning on Buy Box prices for coffee, where we obtain mostly insignificant results in the case of 3P sellers, and a positive coefficient for a lag of 48 weeks in the case of Amazon; see Appendix A.8. This is consistent with evidence showing that the Buy Box algorithm is very price elastic (Chen et al., 2016; Lee and Musolff, 2025; Breslin, 2025).

Figure 9: Pass-through estimates from distributed-lag regressions (Table 3)



(a) Coffee.

(b) Cocoa.

Table 3: Pass-through results: distributed lag regressions

Specification	K (lags)	Sum of Coefficients	Standard Error	T-statistic	Observations
Coffee:					
Third	6	0.18	0.06	3.11	1559169
Third	12	0.30	0.07	4.08	1559169
Third	24	0.44	0.09	4.67	1559169
Third	48	0.69	0.13	5.30	1559169
Amazon	6	-0.03	0.14	-0.20	288299
Amazon	12	-0.18	0.18	-0.97	288299
Amazon	24	-0.18	0.28	-0.63	288299
Amazon	48	0.52	0.38	1.37	288299
Cocoa:					
Third	6	0.16	0.08	1.87	149551
Third	12	0.30	0.10	3.09	149551
Third	24	0.40	0.18	2.18	149551
Third	48	0.97	0.20	4.92	149551
Amazon	6	0.10	0.24	0.42	10996
Amazon	12	0.28	0.38	0.72	10996
Amazon	24	-1.32	0.46	-2.88	10996
Amazon	48	-1.57	0.72	-2.19	10996

Notes: Each row reports the cumulative pass-through estimate, $\sum_{k=0}^{K} \beta_k$, from the distributed-lag specification in equation 5, where K denotes the number of weekly lags of the commodity-price change (cocoa or coffee, respectively) included in the regression. Regressions are estimated separately for 3P seller prices (*Third*) and Amazon prices (*Amazon*) using the baseline levels specification. The dependent variable is the weekly change in product price per gram, and the regressors are the contemporaneous and lagged weekly changes in the commodity price per gram. The sample is a weekly ASIN panel of products from January 2021 to March 2025. Standard errors are two-way cluster-robust. T-statistics are computed using these standard errors.

4.2.3 Discussion

These results are in line with our theory model: While 3P sellers appear to pass through cost shocks (first partially and almost fully with longer lags), Amazon tends to either remain unresponsive to marginal cost changes or, counterintuitively, lower prices following an increase in commodity prices, at least in the time period considered.

On one hand, these results could stem from the advantage that Amazon has, as it does not need to pay itself any commission fee. However, there are admittedly other explanations for this phenomenon. For example, Amazon may have a long-term objective to strengthen its reputation for offering consistently low prices (see Cabral and Xu (2021) or Cavallo, Cavallo and Rigobon (2014) for evidence of such effects), thereby suppressing its margins in the short run. It is also possible that Amazon's broader portfolio and revenue streams can enable it to cross-subsidize specific items, selling them below costs or very small margins, or that it is more efficient.

5 Conclusion

This paper identifies a distinct pricing mechanism on dual-mode e-commerce platforms: because the platform both earns commissions on 3P sales and retails on its own marketplace, its pricing incentives differ from those of 3P sellers. In a stylized model, this generates two key implications: platform retail presence can either soften or intensify price competition, and common cost shocks need not be passed through symmetrically across seller types. Using data from Amazon U.S., we document reduced-form evidence consistent with both implications. 3P prices tend to fall (rise) less strongly following Amazon entry (exit) compared to 3P seller entry (exit); furthermore, 3P sellers appear to pass through commodity cost shocks more strongly than Amazon. These findings highlight a novel channel through which dual-mode operation may shape competition on e-commerce platforms.

Our results are relevant not only for our understanding on price competition on dual-mode platforms, but also in light of recent antitrust cases that focus on pricing practices and restraints on 3P prices in Germany and the U.S. (Federal Trade Commission, 2024; Bundeskartellamt, 2025). In ongoing work, we develop a structural extension that embeds the revenue-sharing vs. cost-saving mechanism in a nested-logit demand system over seller-ASIN varieties, allowing for quantitative analysis of self-preferencing, pass-through, and welfare under counterfactual platform regulation.

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A Empirical Appendix

A.1 Commodity Price Data

Cocoa beans. The cocoa bean pricing data we use stems from the International Cocoa Organization (ICCO) and is available via www.icco.org/statistics. We use the ICCO daily price for cocoa beans in US$, defined as "the average of the quotations of the nearest three active futures trading months on ICE Futures Europe (London) and ICE Futures US (New York) at the time of London close". We refer the interested reader to the website for further information.

Arabica coffee beans. Daily pricing data on Arabica coffee beans are obtained from Trading Economics, available on tradingeconomics.com/commodity/coffee. The data corresponds to Arabica Coffee C Futures contracts that trade on the Intercontinental Exchange (see https://www.ice.com/products/15/Coffee-C-Futures), and is the global benchmark for Arabica Coffee.

For both price series, we create weekly average prices by averaging across all observed prices in a given week. Figure 7 shows the weekly time series of both prices.

A.2 Data Collection from Keepa and Creation of Daily Product Panel

As a first step, we use Keepa's "Product Finder" tool to arrive at a list of ASINs related to a given product category (e.g., Coffee or Men's T-Shirts). Table A.1 shows the query parameters used for each of the products that were queried. Note that a product (ASIN) on Amazon can belong to one *Root Category*, and to multiple *Browse Nodes*[27]. The latter are called *Subcategories* on the Keepa website. We configure the Keepa Product Finder filters so that only subcategories directly related to our target product (e.g., "Ground Coffee") are included, while certain adjacent but irrelevant subcategories (e.g., "Instant Coffee") may explicitly be excluded.[28]

For product categories with over 10,000 products listed on amazon.us, we only collect data on those 10,000 products with the lowest current sales rank or the lowest 180-day average sales rank at the time of scraping, thereby focusing our attention on products that actually generate sales and are (normally) in stock. After obtaining a list of ASINs for a given product category, we query these ASINs one by one using "Product Object" queries[29] through Keepa's Application Programming Interface (API), and thereby obtain product and price information. We query all categories at multiple instances in time in the period of July 2024 through March 2026.

[27]See https://webservices.amazon.com/paapi5/documentation/use-cases/organization-of-items-on-amazon/browse-nodes/browse-nodes-and-items.html (accessed on 25/11/2024). The Browse Node IDs of the subcategories listed in Table A.1 are not displayed for readability reasons, but can be provided upon request.

[28]This way, we can focus on a set of truly highly substitutable products and exclude products that belong to different, dis-similar categories (possibly as a result of misclassification).

[29]See https://keepa.com/#!discuss/t/product-object/116 (last accessed 18/02/2026).

Table A.1: Query parameters used in initial product data collection process on Keepa.

Root Category and Product Category	**Included and excluded subcategories**
Root Category: Grocery & Gourmet Food	
Coffee	· include: Ground Coffee · exclude: Tea; Instant Coffee
Cocoa	· include: Cocoa · exclude: Chocolate Drink Mixes; Hot Chocolate & Malted Drinks; Thanksgiving, Tea & Cocoa
Pasta	· include: Pasta & Noodles · exclude: Fresh Pasta & Sauces; Gnocchi
Root Category: Electronics	
TVs	· include: LED & LCD TVs · exclude: Remote Controls; TV Mounts, Stands & Turntables; Satellite Equipment
USB Hubs	· include: USB Hubs
Micro SD Cards	· include: Micro SD Cards
Root Category: Appliances	
Washing Machines	· include: Washers · exclude: Countertop Dishwashers; Built-in Dishwashers; Portable Dishwashers; Washer Parts & Accessories; Portable Washers
Root Category: Clothing, Shoes & Jewelry	
Men's T-Shirts	· include: Men's T-Shirts · exclude: Tights, Pants & Shorts; Leggings

After collecting the list of ASINs, we exclude products which (either based on manual inspection or based on the variable "type") seem to belong to a different category and are thus misclassified. For products in the Pasta category, we exclude products marked with terms like "buckwheat", "rice" or "gluten-free", so as to focus on products made from wheat; for products in the coffee category, we exclude chicory or instant coffee sticks, for instance. Within a given category, we only use data on products that are ever on sale on `amazon.com` and thus have a price indicated at any point since 2021.

The original "Product Object" data downloaded through the Keepa API essentially consists of separate datasets for the lowest "marketplace" offer (which can be Amazon or a 3P seller's price, called NEW in Keepa); the lowest price set by any 3P seller price operating under FBA; the lowest price set by any 3P seller price operating under FBM; Amazon's price; and the price of the seller whose offer is displayed in the Buybox. One row in this original data corresponds to a given price change at a given moment in time (a price becoming unavailable or available corresponds to a "change" as well). We define the lowest 3P price (NEW_third) as the marketplace price (NEW) whenever it differs from the Amazon price, and as the minimum of the lowest FBA and FBM prices otherwise.

Adjustment for occasionally occurring delayed Amazon price recording. The product-level price data from Keepa occasionally exhibits a timing discrepancy in which Amazon's offer is detected with a short lag. Specifically, when a previously out-of-stock Amazon offer returns to the marketplace, Keepa's system may register the resulting change in the marketplace price (NEW) and seller count (COUNT_NEW) before identifying the Amazon offer itself. During this interim period – lasting only a few days – the AMAZON price field remains missing even though the marketplace price already reflects Amazon's presence, which would cause the lowest 3P price (NEW_third) to be incorrectly attributed. We confirmed this data-collection artifact directly with the Keepa team[30]. To address it, we apply a conservative correction procedure that detects instances in which (i) the seller count increases by exactly one, (ii) the marketplace price simultaneously decreases, (iii) the Amazon price is not yet recorded, (iv) no FBA or FBM price equals the marketplace price on the affected days (ruling out the possibility that the price drop is attributable to a 3P seller), and (v) within seven days the Amazon price appears while the seller count remains *unchanged*. When all four conditions are met, we backfill the Amazon price to the date of the initial seller-count increase. This adjustment affects a small number of product-days (0.001% in the case of Cocoa and 0.002% in the case of coffee, for instance) and ensures that Amazon's entry date is correctly recorded.

Dealing with a small number of apparent "pricing errors". We noticed a certain number of apparent "pricing errors" in the original data, i.e., instances where a product's price jumps up by several multiples of the original price for a period of time. For instance, a package of coffee bearing the ASIN B09B8TP9RQ, typically priced at around \$24.99, is suddenly offered at \$5,889.34 on February 20th to 26th, 2025 – an almost 200-fold increase in its price – before coming back down. These can be due to, e.g., data entry mistakes, or algorithmic malfunctions. We never observe such patterns in the case of Amazon's offers, only for 3P sellers' prices. To detect these upward price deviations in a conservative, but sensible way and exclude them from our dataset upfront, we employ a multi-stage procedure to identify these errors at both the observation and product levels. For each ASIN, we calculate the interquartile range (IQR = Q75 - Q25) of observed prices. An individual price observation is flagged as an outlier if it satisfies all three conditions: (1) Price $>$ Q75 + 30 $\times$ IQR; (2) Price - Q75 $>$ 2 $\times$ median_all, where median_all is the median price across all products; and (3) Price $>$ 2 $\times$ median_product. These criteria jointly ensure that flagged observations represent extreme deviations in both product-specific terms (IQR criterion), absolute terms (preventing spurious flags in low-variance products), and relative terms (substantial deviation from the product's typical price level). While we allow for the exclusion of entire *products* from the sample when outlier prevalence suggests systematic data

[30]See https://keepa.com/#!discuss/t/price-update-frequency-for-amazon/18645 (last accessed 18/02/2026).

quality issues[31], in the case of coffee or cocoa, there are no such systematic data quality issues for any products in our sample. More generally, only around 0.2% of all ASINs are affected by these pricing errors. Finally, in the coffee category, we exclude one brand ("San Marco Coffee") which operated under a dropshipping arrangement until 2024. For this brand, the data show an extreme repricing event, followed by a synchronized exit of Amazon offering any products by that brand. The observed patterns are consistent with a termination in the vendor relationship, which was confirmed to us by the brand, and thus is likely independent from any pass-through of cost shocks.

Sales rank. Given that a product's sales rank is not always recorded on a daily basis, we forward-fill this variable, so that a product's sales rank on a given day reflects the latest sales rank that was recorded on Keepa.

After creating a daily panel of products' prices (recorded at midnight of a given date), we further exclude products if their average 3P sellers' (Amazon's) price (standardized by weight when considering grocery products) across all observed days lies above the 98th, or below the 2nd percentile of all 3P sellers' (Amazon's) daily prices. Very often these products in fact do not belong to the category considered and hence were misclassified, and should hence not be part of our sample.

Defining Amazon "entry" and "exit". We define Amazon entry to be the first day on which Amazon's price variable becomes available, following a day at which it was not available. Likewise, we define exit as the first day on which Amazon's price variable becomes unavailable, following a day of Amazon's price being available.

Building a weekly panel. From a daily panel of products with prices recorded at midnight, we build a weekly panel by taking the average across all observed prices and sales rank. If a price variable is not available on some days of a given week, these take the average across the remaining days on which it is observed. While our analysis covers January 2021 until March 2025, note that, in order to generate the pass-through results, we take lagged variables going back into 2020.

Identifying Amazon private label products. While not core to our analysis, we report the percentage of Amazon private label products in Table 1. We identify these private label products by tagging products whose manufacturer is listed as "365 by Whole Foods Market," "Amazon.com Services, Inc.," "Amazon.com Services LLC.," or "Amazon," as well as products whose brand is listed as "Amazon Fresh" or "Amazon Basics."

[31] Specifically, we remove all observations for a product if either: (1) more than 30% of observed prices are flagged as outliers, or (2) the product has at least 10 outliers and fewer than 50 total price observations.

A.3 Extracting Weight Using the OpenAI API

For our analysis, a precise extraction of product package sizes is crucial. However, package sizes from the relevant variables included by Keepa are missing for many of the products. Instead, package sizes are being accurately described in the product title, but due to nonstandardized formatting, are difficult to extract using standard string-matching methods. To address this, we employ OpenAI's gpt-4o-mini model. The prompt instructs the model to identify and compute information on total weight or volume of the content as indicated in the product title, and is displayed below.

```
You are tasked with extracting the total weight or volume of a grocery
   product from its description. Follow these specific rules:

    1. **Accurate extraction only**: Use only the information explicitly
       provided in the product description. Do not infer or assume any
       details.
    2. **Output format**: Provide the result as ‘number|unit‘.
       - If the description includes a total weight or volume (including
          multipacks), calculate the combined total if necessary.
       - Units like ounces (oz), grams (g), pounds (lb), kilograms (kg),
          liters (L), or milliliters (ml) should be preserved as stated.
    3. **No weight or volume available**: If no weight or volume
       information is provided, return ‘NA|NA‘.

    **Examples**:

    - **Input**: "Starbucks Caffe Verona Coffee, Dark, Ground, 12-Ounce
       Bags (Pack of 3)"
      **Output**: 36|oz

    - **Input**: "Four Sigmatic Think Mushroom Coffee | Organic Ground
       Coffee with Lion’s Mane Mushroom and Chaga Mushroom | 12oz Bag"
      **Output**: 12|oz

    - **Input**: "AmazonFresh Go For The Bold Ground Coffee, Dark Roast"
      **Output**: NA|NA

    - **Input**: "AmazonFresh French Vanilla Flavored Coffee, Ground,
       Medium Roast, 500g"
      **Output**: 500|g
```

```
Now, extract the total weight or volume from the following description
    :

---
{text}
---
```

To guarantee consistency and reproducibility, we set the temperature parameter to 0. We moreover include a system message – "role": "system", "content": "You are an expert at analyzing grocery product descriptions and extracting accurate weight or volume information. You strictly follow instructions, avoid assumptions, and format answers as specified." – which primes the model as a domain expert and ensures it adheres rigorously to the extraction protocol without introducing unwarranted assumptions.

Table A.2 displays the resulting distribution of weights across ASINs, showing that we are able to recover the product weight for a vast majority of products, and showing that the median package size is relatively standard.

Table A.2: Distribution of product weights across ASINs in grocery categories

Category	ASINs	ASINs w/ weight available	Mean	Median	SD	P10	P90
Cocoa	1,432	1,393	1137.2	453.6	2628.3	170.5	2268.0
Coffee	14,532	14,282	828.1	453.6	894.2	270.0	2041.2
Pasta	7,709	7,477	1861.2	997.9	2258.4	340.2	5000.0

Note: The table reports ASIN-level weight statistics (in grams) for Cocoa, Coffee, and Pasta products.

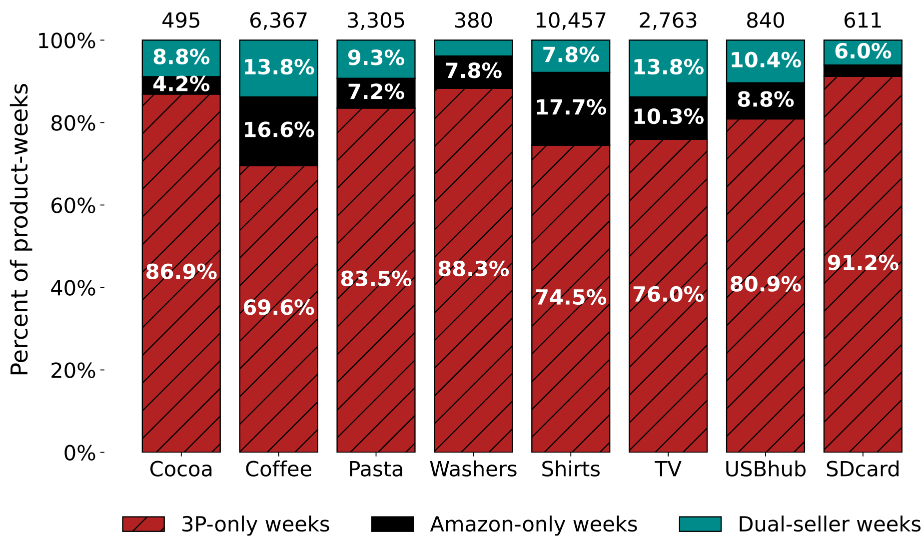
A.4 What Determines Amazon's Decision to Sell a Product as a Retailer?

As Figure 2 shows, the presence of Amazon as a seller (in terms of number of products) varies across product categories. Figure A.1 mirror the idea that Amazon tends to be active in "popular" products, and Figure A.2 analyzes the extent of competition conditional on Amazon presence. Finally, Table A.3 presents summary statistics focusing only on products which ever experience Amazon presence in our sample.

A.4.1 Survey of Previous Literature

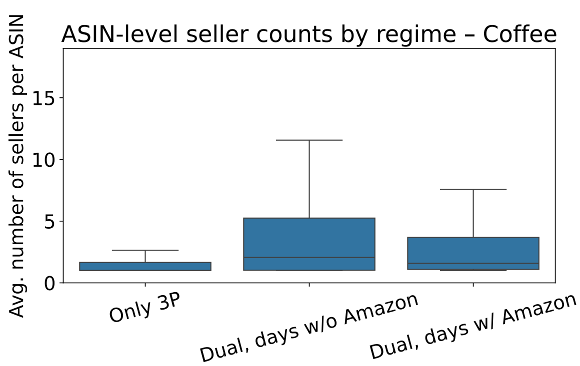
Several academic papers shed light on this question by studying the predictors of Amazon's decision to directly re-sell a product (i.e., to "enter").

Figure A.1: Extent of dual-mode selling across categories, popular products

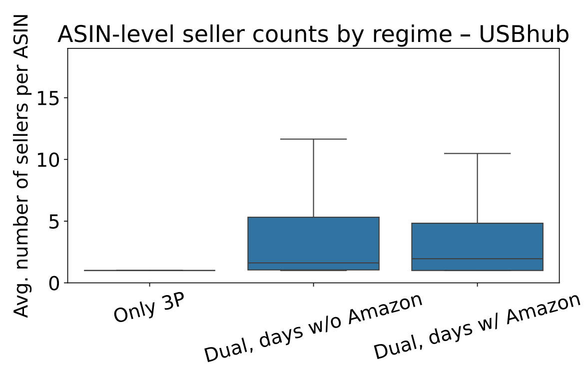


Notes: Bars show, for each category, the share of ASIN-week observations in which the product is sold only by 3P sellers, only by Amazon, or by both. The sample contains **only popular** ASINs (as defined in Section 2.2), observed weekly from January 2021 through December 2024. A product-week is classified as 3P-only (Amazon-only) if at least one 3P (Amazon) offer is recorded and the other side is absent; it is classified as dual-seller if both have offers in the same week.

Figure A.2: Competition by seller regime for exemplary categories



(a) Coffee.



(b) USB Hubs.

Notes: The figure plots, for the Coffee and USB hub categories, the distribution across ASINs of the average daily number of sellers in three regimes. "Only 3P" includes ASINs that are never supplied by Amazon but have at least one 3P seller; for each such ASIN we average the number of sellers over days on which a 3P seller is present. "Dual, days w/o Amazon" restricts to ASINs that are supplied at some point by both Amazon and 3P sellers, but averages the number of sellers only over days when Amazon is absent and at least one 3P seller is present. "Dual, days w/ Amazon" uses the same set of dual ASINs but averages over all days when Amazon is present, regardless of whether a 3P seller is active. (We restrict to days with at least one seller present.)

Hagiu and Wright (2015) use data on DVDs that were sold on Amazon in 2014, and find that products offered by Amazon have lower sales rank, i.e., are more popular, than those offered by

Table A.3: Summary of sample by category, restricted to ASINs ever sold by Amazon (ASIN-week panel; 222 weeks)

Category	Products (ASINs)	Avg. seller count (per ASIN-week)	ASIN-weeks with ≥ 2 sellers	With reviews (across ASINs)	Amazon private label (across ASINs)
Cocoa	160	4.8	62.9%	92.5%	1.9%
Coffee	4,338	3.1	41.0%	97.8%	1.0%
Pasta	1,432	3.7	59.7%	91.3%	2.0%
Washers	93	3.7	45.3%	74.2%	0.0%
TV	1,816	3.5	40.9%	79.2%	0.0%
Shirts	3,365	1.2	7.4%	72.4%	0.0%
USB hubs	879	5.8	36.6%	50.2%	3.3%
SD cards	205	5.3	65.2%	88.3%	2.0%

Notes: Restricted to ASINs where Amazon retail lists the product (`AMAZON` non-missing) in at least one ASIN-week. We track sampled ASINs daily from January 1, 2021 to March 31, 2025 and collapse the data to the ASIN-week level. The average seller count corresponds to the average number of sellers listing an ASIN in a week, computed over ASIN-weeks with at least one observed seller, and includes Amazon retail when present. "With reviews" and "Amazon private label" are product-level shares. The review indicator is measured from the latest observed product page and is therefore time-invariant within the sample.

3P sellers.

Zhu and Liu (2018) employs data from 2013 and 2014 and finds that Amazon is more likely to begin re-selling products that experience higher sales and have better reviews, and which are not yet using Amazon's fulfillment services.

A working paper by Crawford et al. (2022) uses proprietary data on prices and revenues in Amazon's Home & Kitchen category. The paper focuses on Amazon's decision to begin selling products for which only a single 3P merchant supplied that product in the first four weeks of the product's existence. The authors find that Amazon starts re-selling products that experience high demand growth and where competition is low (i.e., few active sellers are present). This is true both relative to products which Amazon does *not* enter, as well as relative to the entry decisions by big (high-revenue) 3P sellers. Entry by Amazon occurs even if a large or efficient 3P seller is present. Amazon is also more likely to enter when 3P listings exist but generate no sales, suggesting the presence of low-quality or capacity-constrained sellers. The authors argue that these patterns suggest the internalization of platform externalities as a possible motive for Amazon's entry decisions. Among those externalities are expanding product variety, ensuring product availability, and mitigating seller market power.

Concerning Amazon's presence as a re-seller more generally, the authors find that nearly half of the revenue from products in which Amazon is a reseller comes from cases of *de novo* entry – products that had not been sold by any 3P seller before Amazon's entry as a reseller. Amazon's private label products play a minor role in the Kitchen & Home category, accounting for only 0.3% of total revenues. Amazon is present in products accounting for 39% of revenues in the

Home & Kitchen category.

The findings by Bennati (2026), who studies the market for headphones and uses sales estimates stemming from the market intelligence company AmzScout, are very much in line with this. The author's findings suggest that 3P sellers concentrate on products in the long tail, whereas Amazon re-sells products that are in higher demand and of higher quality.

A.4.2 Amazon Entry in Our Data

Table A.4: Correlates of whether a product is ever supplied by Amazon, Coffee products only.

	Dependent variable:	
	$\mathbb{1}\{$product ever sold by Amazon$\}$	
1{popular}	0.223^{**}	0.162^{***}
	(0.090)	(0.047)
1{has any reviews}	0.175^{***}	0.025
	(0.061)	(0.033)
avg number of 3P sellers	0.008	0.007^{***}
	(0.013)	(0.003)
package size (weight, in kg)	$1.002\,e^{-8}$	$8.316\,e^{-10}$
	$(1.913\,e^{-8})$	(5.273ee−9)
product was ever stocked out	0.164^{***}	0.014
	(0.056)	(0.031)
Manufacturer FE		✓
Brand FE		✓
Observations	12,759	12,664
Adjusted R^2	0.076	0.788
SEs clustered at manufacturer level.	$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$	

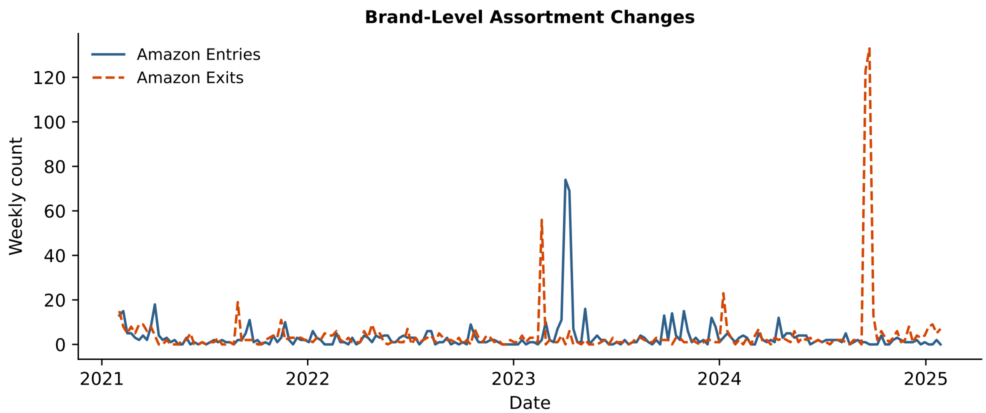
Notes: Cross-sectional dataset was created by beginning with a daily panel ranging from January 1st 2021 to March 31 2025, and collapsing it to the cross-sectional level. 1{popular} is the popularity measure defined in Section 2.2, indicating the proportion of days on which a product's sales rank is below the median sales rank across products, and hence varies between 0 and 1. The average number of 3P sellers is defined as an average across all days on which a product is available for sale. The coefficients from a linear regression are shown, and each observation is a product.

Taking the example of coffee, we find that, in the cross-section, the brand and the manufacturer are main factors correlated with Amazon's entry patterns. Controlling for brand and manufacturer, we moreover find that products where Amazon is a reseller at any point during our period of observation are likely to derive more sales, and are likely to feature more other sellers, as can be seen in Table A.4. The finding that more competition by 3P sellers is correlated with Amazon re-selling the product could in fact be driven by unobserved (by the econometrician) demand shocks, which might cause entry by both Amazon and other merchants. Hence, our measure of "high competition" might in fact be a proxy for "high demand". (Of course, this result can by no means be interpreted causally, and is purely descriptive.)

In the time series, interestingly we find that, across our product categories, the overall number of products re-sold by Amazon has been constant over the years, while the number of products re-sold by 3P sellers has increased over time (not shown in paper).

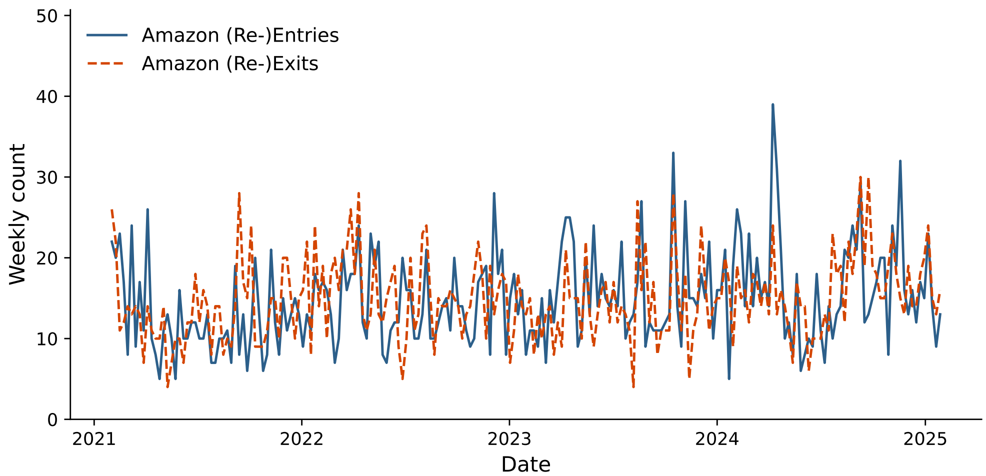
A.5 Analysis of Recurring Amazon Presence Within ASIN

Figure A.3: Amazon assortment change on the brand level: case of "The Coffee Fool"



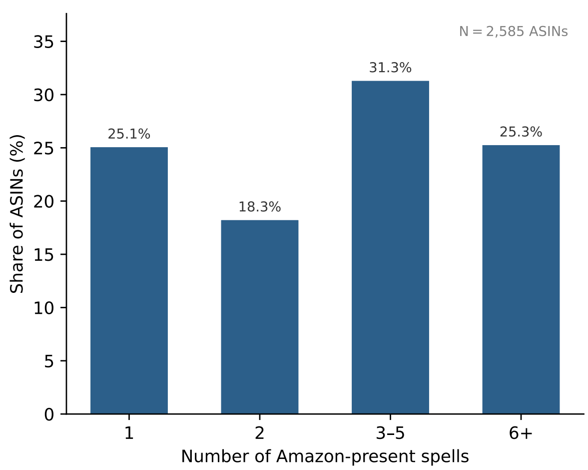
Notes: To illustrate that Amazon sometimes enters or exits systematically many ASINs of a particular brand, this Figure shows the weekly number of (re-)entries and (re-)exits of Amazon over time in the coffee category, focusing on the brand "The Coffee Fool" only. We only count a re-entry (re-exit) as such if they are followed by a spell of presence (absence) of 5 days or more.

Figure A.4: Recurring Amazon entries and exits over time: Pasta category

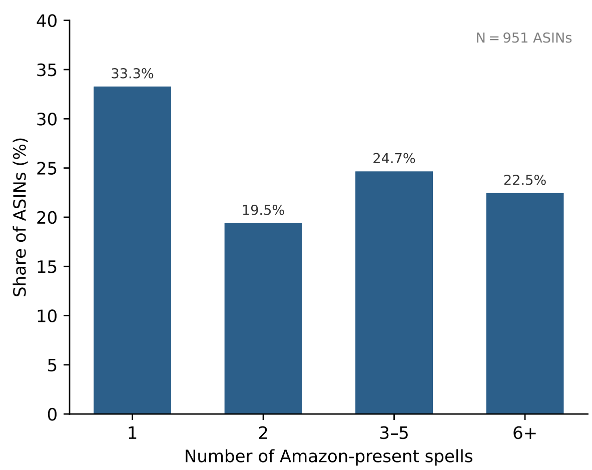


Notes: The Figure shows the analogous to Figure 3, but this time focusing on our data on Pasta. The lines indicate the weekly number of re-exits and re-entries by Amazon over time for the Pasta category. We exclude brands that account for more than 20% of aggregate Amazon entry or exit flows in the ten highest-activity weeks (two brands in the case of pasta), as these spikes reflect systematic brand-level assortment changes rather than product-level entry or exit decisions. We only count a re-entry (re-exit) as such if they are followed by a spell of presence (absence) of 5 days or more (this is robust to other cutoffs).

Figure A.5: Recurring spells of Amazon presence within ASIN



(a) Ground coffee category.



(b) Wheat pasta category.

Notes: The bars show the percentage of ASINs within the indicated category with 1, 2, 3–5, or 6 or more "spells," i.e., contiguous runs of days during which Amazon lists a price for the product. We exclude brands that account for more than 20% of aggregate Amazon entry or exit flows in the ten highest-activity weeks (two brands in the case of pasta), as these spikes reflect systematic brand-level assortment changes rather than product-level entry or exit decisions. We further restrict to spells lasting at least five consecutive days to exclude brief stockout episodes that appear as transient Amazon absences in the raw data. N denotes the number of ASINs with at least one qualifying spell. The sample covers products in the ground coffee and wheat pasta categories and runs from January 2021 through March 2025.

Table A.5: Descriptives: Seller regimes (FBA as 3P price)

	Obs.	Share	N (ASINs)
Panel A: ASIN-day regime counts			
3P only	695,734	0.251	1,393
Amazon only	851,903	0.308	1,628
Both	316,128	0.114	1,425
Neither	906,321	0.327	1,779
Total	2,770,086	1.000	1,786

	Mean	Median	P10	P90
Panel B: Within-ASIN share of days in regime				
3P only	0.251	0.121	0.000	0.716
Amazon only	0.308	0.273	0.001	0.758
Both	0.114	0.047	0.000	0.324
Neither	0.327	0.237	0.029	0.755

	N (spells)	Mean duration	Median duration
Panel C: Regime persistence (spell duration in days)			
3P only	16,350	42.55	11.00
Amazon only	14,488	58.80	6.00
Both	13,165	24.01	4.00
Neither	18,104	50.06	6.00
Regime transitions per ASIN			
Mean	33.77		
Median	26.00		

Notes: Coffee category; ASIN-day panel (midnight prices), January 2021–March 2025. ASINs are restricted to those satisfying the event-study screen used in this script: at least 10 consecutive days with both Amazon and FBA non-missing, and at least 10 consecutive days with only the third-party price observed (Amazon missing). Regimes are defined by price availability of Amazon and FBA. "3P only": 3P price observed and Amazon missing. "Amazon only": Amazon observed and 3P missing. "Both": both observed. "Neither": neither observed.

Table A.6: Descriptives on Amazon retail presence spell duration among switcher ASINs (FBA-active days)

	N	Mean	Std. Dev.	Median	P10	P90
Panel A: Spell-level distribution (spell duration in days)						
Amazon present	9,472	31.89	90.13	3.00	1.00	83.90
Amazon absent	9,722	66.23	158.36	9.00	1.00	176.00
Panel B: Per-ASIN number of spells						
Amazon present	1,261	7.51	11.41	3.00	1.00	17.00
Amazon absent	1,261	7.71	11.49	4.00	1.00	17.00
Panel C: Per-ASIN mean spell duration (days)						
Amazon present	1,261	65.14	92.35	34.00	4.21	157.50
Amazon absent	1,261	146.42	200.84	62.33	6.00	421.00

Notes: Coffee category; ASIN-day panel (midnight prices), January 2021–March 2025. ASINs are restricted to those satisfying the event-study screen: at least 10 consecutive days with both Amazon and NEW_third non-missing, and at least 10 consecutive days with NEW_third observed and Amazon missing. We further restrict to switcher ASINs in the 3P-active sample (days with FBA observed): at least one day where Amazon is present and one day where it is absent. A spell is a maximal run in which Amazon's presence status is unchanged. Panel A reports spell-duration distributions for Amazon-present and Amazon-absent spells. Panel B reports the per-ASIN number of spells (separately for Amazon-present and Amazon-absent spells). Panel C reports each ASIN's mean spell duration (separately for Amazon-present and Amazon-absent spells).

Table A.7: Logit model: Predictors of clean Amazon entry (3P-only days at risk, using FBA prices)

Model:	(1) Baseline (1)	(2) + 3P price (2)	(3) + Sales rank (3)	(4) + lagged changes (4)
Num. 3P sellers $(t-1)$	0.0400***	0.0417***	0.0412***	0.0348**
	(0.0140)	(0.0143)	(0.0144)	(0.0161)
weekdayTue	0.4278**			
	(0.1845)			
weekdayWed	0.7976***			
	(0.1777)			
weekdayThu	0.6070***			
	(0.1821)			
weekdayFri	0.5886***			
	(0.1830)			
weekdaySat	0.4593**			
	(0.1864)			
weekdaySun	0.0822			
	(0.1976)			
FBA price $(t-1)$		0.0090	0.0107	0.0228***
		(0.0111)	(0.0107)	(0.0083)
log Sales rank $(t-1)$			-0.0730*	-0.0961**
			(0.0396)	(0.0444)
Δ_3 log Sales rank $(t-1)$				-0.0508
				(0.0701)
Δ Num. 3P sellers $(t-1)$				0.1874***
				(0.0537)
Δ FBA price $(t-1)$				-0.0062
				(0.0167)
Δ FBM price $(t-1)$				0.0163
				(0.0139)
ASIN	Yes	Yes	Yes	Yes
Week	Yes	Yes	Yes	Yes
Observations	227,755	227,755	224,960	151,766

Notes: Clustered (ASIN) standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$. The estimation sample consists of all ASIN-days at risk of entry, that is, every 3P-only day (Amazon absent) for switcher ASINs (ASINs that ever switch between a "dual" and a 3P-only regime). The dependent variable equals one if a baseline qualifying FBA-eligible Amazon entry occurs on the following day and zero otherwise. At-risk days on which Amazon enters but the transition is non-qualifying are dropped. Models are estimated by discrete-time logit. Δ_3 log Sales rank $(t-1)$ is the three-day change in the natural log of the Amazon sales rank measured at $t-1$, i.e. $\log(\text{rank}_{t-1}) - \log(\text{rank}_{t-4})$; it approximates the three-day percentage change in sales rank. Because sales rank is inversely related to units sold, a positive value indicates a deterioration (decline) in the product's relative sales performance.

Table A.8: Logit model: Predictors of clean Amazon exit (both-regime days at risk, using FBA prices)

Model:	(1) Baseline (1)	(2) + Prices (2)	(3) + Sales & buy box (3)	(4) + Amazon higher + lagged changes (4)
Num. 3P sellers $(t-1)$	0.0455**	0.0454**	0.0464**	0.0417**
	(0.0194)	(0.0193)	(0.0188)	(0.0205)
weekdayTue	0.1438			
	(0.1663)			
weekdayWed	0.2188			
	(0.1729)			
weekdayThu	-0.0301			
	(0.1750)			
weekdayFri	0.1057			
	(0.1742)			
weekdaySat	0.2361			
	(0.1659)			
weekdaySun	0.0168			
	(0.1863)			
Amazon price $(t-1)$		-0.0271	-0.0227	-0.0152
		(0.0188)	(0.0196)	(0.0218)
FBA price $(t-1)$		0.0011	-0.0045	-0.0031
		(0.0104)	(0.0119)	(0.0120)
log Sales rank $(t-1)$			0.1307**	0.1331**
			(0.0578)	(0.0646)
Amazon wins buy box $(t-1)$			0.2608	0.4202*
			(0.1766)	(0.2358)
Δ_3 log Sales rank $(t-1)$				0.0384
				(0.1449)
Amazon $>$ 3P price $(t-1)$				0.1232
				(0.2079)
Δ Num. 3P sellers $(t-1)$				0.1239
				(0.0857)
Δ FBA price $(t-1)$				-0.0021
				(0.0278)
Δ FBM price $(t-1)$				0.0024
				(0.0062)
Δ Amazon price $(t-1)$				-0.0080*
				(0.0046)
ASIN	Yes	Yes	Yes	Yes
Week	Yes	Yes	Yes	Yes
Observations	73,220	73,220	72,387	46,037

Notes: Clustered (ASIN) standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$. The estimation sample consists of all ASIN-days at risk of exit, that is, every day with Amazon present for switcher ASINs (ASINs that ever switch between a "dual" and a 3P-only regime). The dependent variable equals one if a baseline qualifying FBA-eligible Amazon exit occurs on the following day and zero otherwise. At-risk days on which Amazon exits but the transition is non-qualifying are dropped. Models are estimated by discrete-time logit. Δ_3 log Sales rank $(t-1)$ is the three-day change in the natural log of the Amazon sales rank measured at $t-1$, i.e. $\log(\text{rank}_{t-1}) - \log(\text{rank}_{t-4})$; it approximates the three-day percentage change in sales rank. Because sales rank is inversely related to units sold, a positive value indicates a deterioration (decline) in the product's relative sales performance.

A.6 Analysis of Prices: Further Findings

Differences in price levels. To make a precise comparison of 3P sellers' and Amazon's prices in product-weeks in which both are selling, Tables A.9 to A.13 display the results from a regression of logged prices on an indicator for Amazon, with ASIN and year-week fixed effects. It becomes apparent that, if anything, Amazon tends to set a lower price than 3P sellers, especially in the coffee category.

Price adjustment frequency. Tables A.14 and A.15 show that, conditional on both parties selling a given ASIN, there are no substantial differences in price adjustment frequency (if anything, Amazon's price is more likely to display a higher week-to-week price change than the lowest 3P seller's price).

Table A.9: Price differences for identical products: Coffee

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	-0.2652***	-0.1860***	-0.4245***	-0.2609***	-0.1910***	-0.4266***
	(0.0072)	(0.0082)	(0.0069)	(0.0078)	(0.0088)	(0.0075)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	286,808	202,772	273,438	247,398	177,311	236,672
Adjusted R^2	0.88493	0.92332	0.91610	0.88489	0.92270	0.91726

Notes: The unit of observation is an ASIN–week in the Coffee category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. "Popular products" are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table A.10: Price differences for identical products: Cocoa

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	-0.0395	-0.0568*	-0.2676***	-0.0339	-0.0449	-0.2808***
	(0.0469)	(0.0306)	(0.0351)	(0.0529)	(0.0319)	(0.0382)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	13,504	10,371	12,917	11,730	9,195	11,231
Adjusted R^2	0.90274	0.93924	0.93063	0.87096	0.91966	0.90680

Notes: The unit of observation is an ASIN–week in the Cocoa category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. "Popular products" are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table A.11: Price differences for identical products: T-Shirts

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	0.0002	0.0027***	-0.0832***	-0.0006	0.0039***	-0.1621***
	(0.0014)	(0.0006)	(0.0114)	(0.0014)	(0.0006)	(0.0226)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	282,208	270,711	157,649	212,278	208,892	110,913
Adjusted R^2	0.81860	0.86944	0.78345	0.82006	0.86495	0.79109

Notes: The unit of observation is an ASIN–week in the Men's T-Shirts category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. "Popular products" are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table A.12: Price differences for identical products: SD Cards

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	-0.0309	-0.0006	-0.1016***	-0.0150	0.0578	-0.0895**
	(0.0192)	(0.0243)	(0.0226)	(0.0298)	(0.0376)	(0.0360)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	20,218	15,719	19,126	10,620	7,963	9,968
Adjusted R^2	0.91216	0.92688	0.91114	0.89789	0.91437	0.89695

Notes: The unit of observation is an ASIN–week in the SD Cards category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. "Popular products" are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table A.13: Price differences for identical products: USB Hubs

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	0.0028	0.0289	-0.0914***	0.0046	0.0542	-0.0967***
	(0.0170)	(0.0211)	(0.0146)	(0.0214)	(0.0336)	(0.0178)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	36,666	27,215	31,872	19,786	14,702	16,881
Adjusted R^2	0.92494	0.93599	0.93954	0.92581	0.93043	0.94510

Notes: The unit of observation is an ASIN–week in the USB Hubs category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. "Popular products" are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table A.14: Amazon versus 3P weekly price adjustment frequency: Cocoa category

Sample	Joint	Joint	Broad	Broad
Dependent Variables:	Adj. indicator (>0.5%)	Adj. indicator (>1%)	Adj. indicator (>0.5%)	Adj. indicator (>1%)
Model:	(1)	(2)	(3)	(4)
Amazon indicator	0.0180	0.0090	0.0175	0.0055
	(0.0330)	(0.0310)	(0.0321)	(0.0296)
ASIN	Yes	Yes	Yes	Yes
Week	Yes	Yes	Yes	Yes
Observations	12,470	12,470	160,556	160,556
R^2	0.13828	0.12788	0.25566	0.22198

Notes: Dependent variable equals one if the absolute weekly log price change exceeds the threshold shown in each column (0.5% and 1% respectively), and zero otherwise. Joint sample restricts to ASIN-week observations with non-missing Amazon and third-party prices in both t and $t-1$; broad sample uses all seller-week observations with non-missing own prices in t and $t-1$. Weekly data, January 2021 to March 2025. All columns include ASIN and week fixed effects. Clustered (ASIN) standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

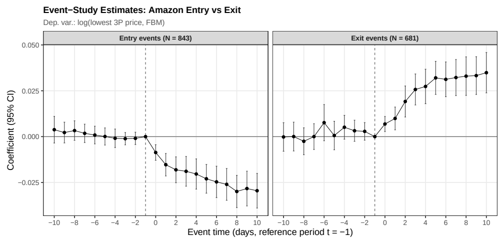
Table A.15: Amazon versus 3P weekly price adjustment frequency: Coffee category

Sample	Joint	Joint	Broad	Broad
Dependent Variables:	Adj. indicator (>0.5%)	Adj. indicator (>1%)	Adj. indicator (>0.5%)	Adj. indicator (>1%)
Model:	(1)	(2)	(3)	(4)
Amazon indicator	0.0203***	-0.0031	0.0551***	0.0254***
	(0.0071)	(0.0065)	(0.0064)	(0.0057)
ASIN	Yes	Yes	Yes	Yes
Week	Yes	Yes	Yes	Yes
Observations	263,530	263,530	1,841,761	1,841,761
R^2	0.10842	0.10250	0.26324	0.22855

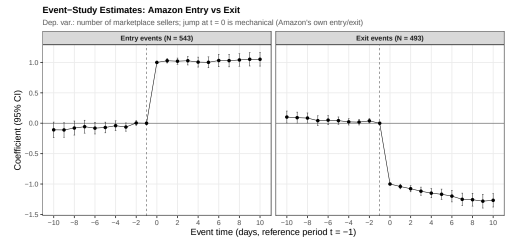
Notes: Dependent variable equals one if the absolute weekly log price change exceeds the threshold shown in each column (0.5% and 1% respectively), and zero otherwise. Joint sample restricts to ASIN-week observations with non-missing Amazon and third-party prices in both t and $t-1$; broad sample uses all seller-week observations with non-missing own prices in t and $t-1$. Weekly data, January 2021 to March 2025. All columns include ASIN and week fixed effects. Clustered (ASIN) standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

A.7 Amazon Retail Presence and 3P Price Levels: Additional Results

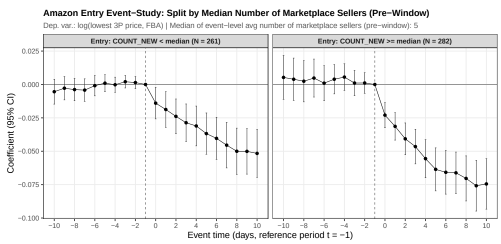
Figure A.6: Event study of entry and exit by Amazon as a seller into a given ASIN: further results



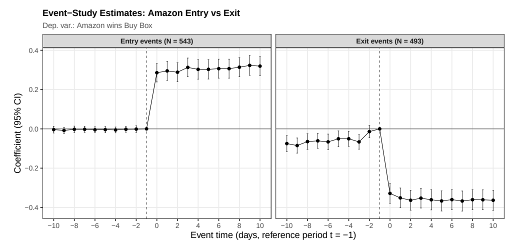
(a) Lowest 3P seller's price under FBM as outcome.



(b) Total number of marketplace sellers.



(c) Split by competitive environment.



(d) Likelihood of Amazon buy box wins.

Tables A.17 and A.18 show the results from a more "naive", but more standard, two-way fixed effects (TWFE) specification using data at the daily level. We run separate regressions for each product category. For product i on day t, we estimate:

$$\log FBA_{it} = \beta \cdot \mathbb{I}(\text{Amazon}_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{6}$$

where, as in the main text, FBA_{it} denotes the lowest price offered by 3P sellers under the FBA model of ASIN i on day t (FBA_{it} is defined analogously). $\mathbb{I}(\text{Amazon}_{it})$ is an indicator taking the value 1 if Amazon is active as a seller for product i at time t. The inclusion of product fixed effects (μ_i) and time fixed effects (λ_t) absorbs time-invariant product characteristics (e.g., quality) and common macro-shocks.

Table A.16: Amazon entry and exit event-studies: log(FBA) and log(FBM) outcomes

Dependent Variables:	log(FBA price)	log(FBM price)	log(FBA price)	log(FBM price)
Specification	Entry, log(FBA)	Entry, log(FBM)	Exit, log(FBA)	Exit, log(FBM)
Model:	(1)	(2)	(3)	(4)
event_time = -10	0.0001	0.0038	0.0021	-0.0002
	(0.0048)	(0.0037)	(0.0040)	(0.0040)
event_time = -9	0.0007	0.0022	0.0026	3.69×10^{-5}
	(0.0046)	(0.0029)	(0.0041)	(0.0040)
event_time = -8	-0.0006	0.0033	0.0010	-0.0025
	(0.0045)	(0.0027)	(0.0045)	(0.0037)
event_time = -7	0.0005	0.0018	-0.0003	1.16×10^{-5}
	(0.0042)	(0.0026)	(0.0043)	(0.0036)
event_time = -6	7.58×10^{-5}	0.0009	0.0034	0.0076
	(0.0038)	(0.0025)	(0.0044)	(0.0050)
event_time = -5	0.0025	7.86×10^{-5}	0.0018	0.0006
	(0.0032)	(0.0024)	(0.0037)	(0.0040)
event_time = -4	0.0028	-0.0009	-0.0016	0.0051
	(0.0029)	(0.0026)	(0.0036)	(0.0033)
event_time = -3	0.0015	-0.0011	0.0017	0.0032
	(0.0027)	(0.0017)	(0.0032)	(0.0029)
event_time = -2	0.0013	-0.0009	-0.0013	0.0029
	(0.0023)	(0.0017)	(0.0024)	(0.0024)
event_time = 0	-0.0186***	-0.0086***	0.0106***	0.0069***
	(0.0038)	(0.0022)	(0.0033)	(0.0021)
event_time = 1	-0.0253***	-0.0153***	0.0156***	0.0099***
	(0.0042)	(0.0031)	(0.0038)	(0.0032)
event_time = 2	-0.0326***	-0.0181***	0.0183***	0.0192***
	(0.0045)	(0.0036)	(0.0041)	(0.0043)
event_time = 3	-0.0379***	-0.0189***	0.0242***	0.0257***
	(0.0049)	(0.0041)	(0.0046)	(0.0043)
event_time = 4	-0.0438***	-0.0204***	0.0256***	0.0273***
	(0.0054)	(0.0042)	(0.0048)	(0.0048)
event_time = 5	-0.0507***	-0.0230***	0.0288***	0.0320***
	(0.0058)	(0.0040)	(0.0050)	(0.0046)
event_time = 6	-0.0536***	-0.0247***	0.0286***	0.0312***
	(0.0058)	(0.0044)	(0.0053)	(0.0048)
event_time = 7	-0.0562***	-0.0260***	0.0290***	0.0322***
	(0.0060)	(0.0044)	(0.0051)	(0.0050)
event_time = 8	-0.0606***	-0.0299***	0.0265***	0.0330***
	(0.0063)	(0.0045)	(0.0061)	(0.0054)
event_time = 9	-0.0635***	-0.0283***	0.0267***	0.0333***
	(0.0067)	(0.0048)	(0.0060)	(0.0053)
event_time = 10	-0.0635***	-0.0295***	0.0287***	0.0349***
	(0.0068)	(0.0048)	(0.0061)	(0.0056)
Event	✓	✓	✓	✓
Observations	11,403	17,703	10,353	14,301
Number of Events	543	843	493	681
Number of ASINs	414	605	354	489
R^2	0.97950	0.98778	0.98561	0.98530

Notes: Two-way clustered (ASIN × date) standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$. Columns (1) and (2) report entry event studies; columns (3) and (4) report exit event studies. Dependent variables are log(lowest 3P seller price under FBA) in columns (1) and (3) and log(lowest 3P seller price under FBM) in columns (2) and (4). Coefficients are event-time indicators relative to $t = -1$ (omitted period), as specified in equation 3. Entry and exit columns, respectively, use the sample of all qualifying "clean" Amazon-entry and Amazon-exit events, as defined in the main text (multiple events per ASIN are possible).

Figure A.7: Event study of entry and exit by Amazon as a seller into a given ASIN: analysis of sales rank around entry/exit.

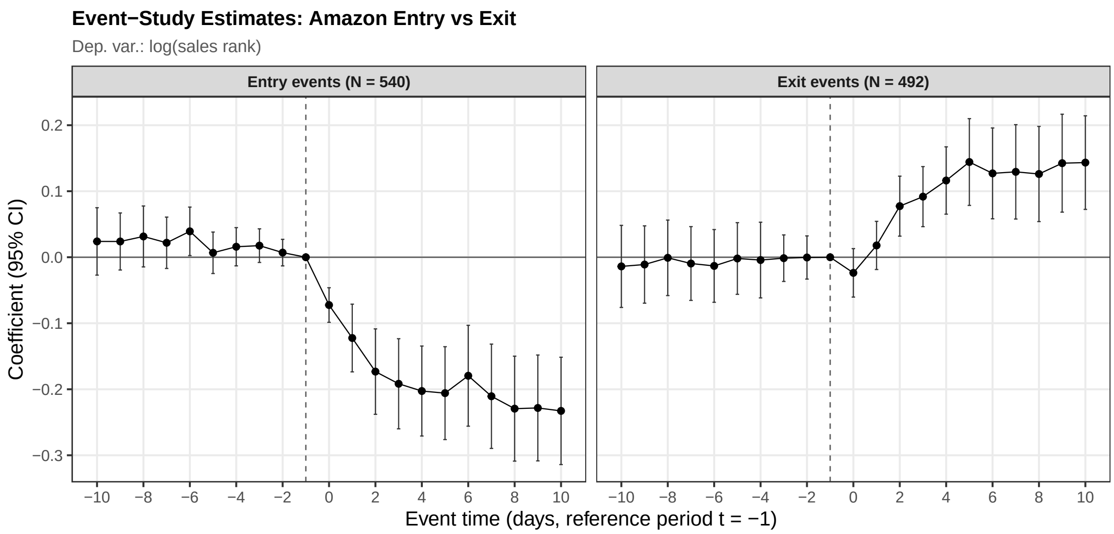
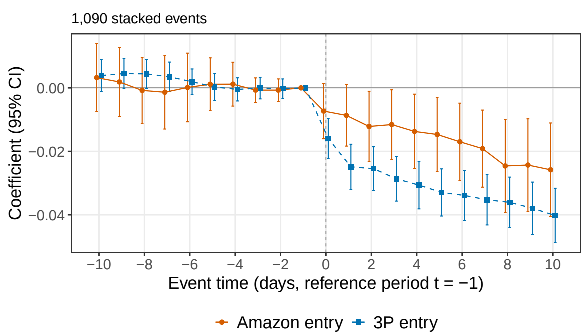
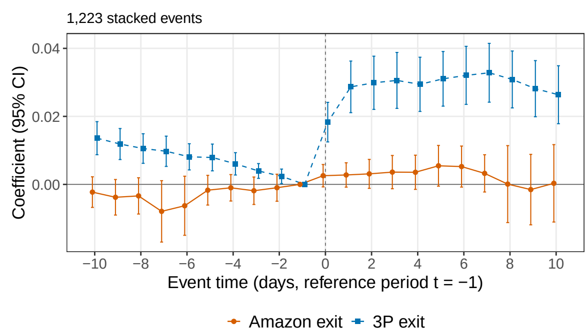


Figure A.8: "Stacked" event study of entry and exit by Amazon and 3P seller; outcome variable: $p^{3p,FBA}$



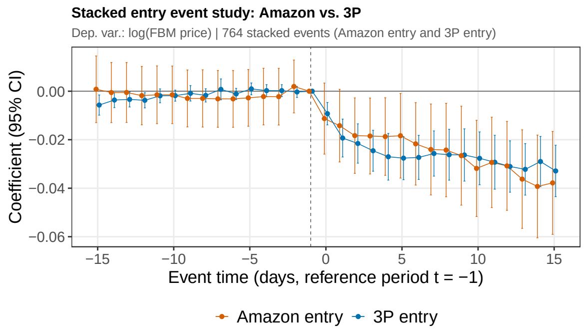
(a) Entry.



(b) Exit.

Notes: Figures show coefficients from estimating equation 4 with the lowest 3P sellers' price under FBA as an outcome variable, as well as 95% confidence intervals computed using two-way clustered standard errors (ASIN and calendar-date). Panel (a) uses entry events; Panel (b) uses exit events. All regressions are estimated using "clean" events in the daily coffee data, as defined earlier.

Figure A.9: "Stacked" event study of entry and exit by Amazon and 3P seller; longer pre-/post windows; outcome variable: $p^{3p,FBM}$



(a) Entry.

(b) Exit.

Notes: Figures show coefficients from estimating equation 4 with the lowest 3P sellers' price under FBM as an outcome variable, as well as 95% confidence intervals computed using two-way clustered standard errors (ASIN and calendar-date). Panel (a) uses entry events; Panel (b) uses exit events. All regressions are estimated using "clean" events in the daily coffee data, as defined earlier.

Table A.17: Amazon's presence and 3P lowest price

Product	Coffee	Cocoa	Pasta	Shirts	TVs	USBhubs
Dependent Variable:	log(FBA)					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
1(Amazon present)	-0.092***	-0.073***	-0.098***	-0.068***	-0.029***	-0.077***
	(0.006)	(0.020)	(0.010)	(0.010)	(0.008)	(0.020)
Fixed-effects						
asin	Yes	Yes	Yes	Yes	Yes	Yes
yearmonthday	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	929,816	59,389	474,405	93,726	127,468	106,130
N ASINs	1,430	71	612	189	395	151
Adjusted R^2	0.90922	0.94750	0.92433	0.77291	0.98274	0.93941

Notes: Sample used: ASIN-day panels (midnight prices), January 2021–March 2025. In each category, ASINs are restricted to those satisfying the event-study screen used in this analysis: at least 10 consecutive days with both Amazon and FBA non-missing, and at least 10 consecutive days with only the 3P price observed (Amazon missing). We do not include SD cards or Washers due to a low number of qualifying ASINs. Clustered (asin) standard-errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

Table A.18: Amazon's presence and 3P lowest price

Product	Coffee	Cocoa	Pasta	Shirts	TVs	USBhubs
Dependent Variable:	log(FBM)					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
1(Amazon present)	-0.030***	-0.053***	-0.055***	-0.018**	-0.046***	-0.067***
	(0.006)	(0.018)	(0.007)	(0.009)	(0.008)	(0.016)
Fixed-effects						
asin	Yes	Yes	Yes	Yes	Yes	Yes
yearmonthday	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	1,487,612	77,678	787,488	172,225	464,300	104,707
N ASINs	1,602	79	761	226	497	117
Adjusted R^2	0.91245	0.95494	0.91918	0.81099	0.96782	0.91219

Notes: Sample used: ASIN-day panels (midnight prices), January 2021–March 2025. In each category, ASINs are restricted to those satisfying the event-study screen used in this analysis: at least 10 consecutive days with both Amazon and FBA non-missing, and at least 10 consecutive days with only the 3P price observed (Amazon missing). We do not include SD cards or Washers due to a low number of qualifying ASINs. Clustered (asin) standard-errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

A.8 Pass-through: additional specifications

Table A.19: Distributed-lag regressions, stacked seller-type panel (common sample), Coffee: Sum of coefficients and standard error

Specification	K (lags)	Sum of Coefficients	Standard Error	T-statistic	Observations
3P	6	0.463	0.287	1.61	264442
3P	12	0.820	0.377	2.18	264442
3P	24	1.085	0.469	2.31	264442
3P	48	2.257	0.808	2.79	264442
Amazon	6	-0.120	0.166	-0.72	264442
Amazon	12	-0.170	0.198	-0.86	264442
Amazon	24	-0.212	0.261	-0.81	264442
Amazon	48	0.266	0.358	0.74	264442
Differential (Amazon - Third)	6	-0.583	0.335	-1.74	264442
Differential (Amazon - Third)	12	-0.990	0.392	-2.52	264442
Differential (Amazon - Third)	24	-1.297	0.464	-2.79	264442
Differential (Amazon - Third)	48	-1.990	0.786	-2.53	264442

Table A.20: Distributed-lag regressions, stacked seller-type panel (common sample), Cocoa: Sum of coefficients and standard error.

Specification	K (lags)	Sum of Coefficients	Standard Error	T-statistic	Observations
3P	6	0.366	0.235	1.56	12470
3P	12	0.631	0.191	3.31	12470
3P	24	-1.088	0.918	-1.19	12470
3P	48	3.431	1.342	2.56	12470
Amazon	6	0.278	0.446	0.62	12470
Amazon	12	0.227	0.666	0.34	12470
Amazon	24	-2.097	0.812	-2.58	12470
Amazon	48	-2.606	1.414	-1.84	12470
Differential (Amazon - Third)	6	-0.089	0.454	-0.20	12470
Differential (Amazon - Third)	12	-0.404	0.739	-0.55	12470
Differential (Amazon - Third)	24	-1.009	1.257	-0.80	12470
Differential (Amazon - Third)	48	-6.037	1.708	-3.53	12470

Figure A.10: Coefficients, stacked specification: Coffee

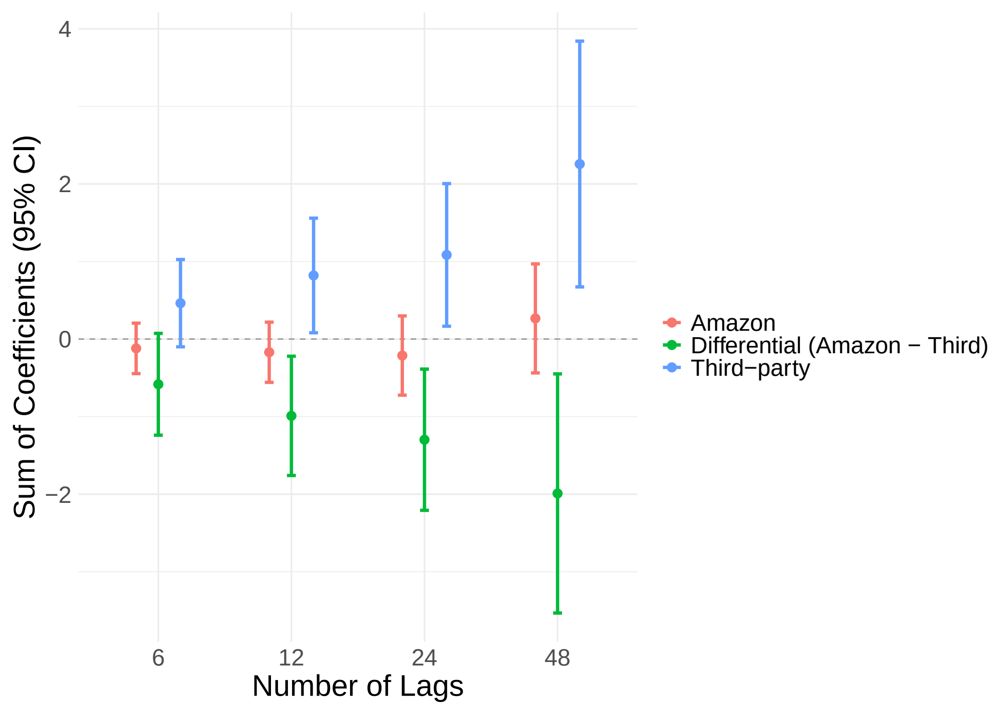


Table A.21: Pass-through: distributed lag regressions using lowest 3P sellers' prices under FBA or FBM, respectively

Specification	K (lags)	Sum of Coefficients	Standard Error	T-statistic	Observations
Coffee:					
FBA	6	0.20	0.06	3.34	769961
FBA	12	0.26	0.08	3.16	769961
FBA	24	0.30	0.10	2.92	769961
FBA	48	0.70	0.13	5.28	769961
FBM	6	0.09	0.05	1.87	1125832
FBM	12	0.20	0.09	2.31	1125832
FBM	24	0.27	0.11	2.52	1125832
FBM	48	0.48	0.14	3.45	1125832
Cocoa:					
FBA	6	0.04	0.06	0.77	58872
FBA	12	0.11	0.08	1.32	58872
FBA	24	0.04	0.15	0.24	58872
FBA	48	0.90	0.15	5.83	58872
FBM	6	-0.08	0.12	-0.69	101916
FBM	12	0.06	0.11	0.53	101916
FBM	24	0.28	0.19	1.53	101916
FBM	48	0.79	0.17	4.66	101916

Notes: Each row reports the cumulative pass-through estimate, $\sum_{k=0}^{K} \beta_k$, from the distributed-lag specification in equation 5, where K denotes the number of weekly lags of the commodity-price change (coffee and cocoa, respectively) included in the regression. Regressions are estimated separately for the lowest 3P seller price under FBM, and the lowest 3P seller price under FBA, using the baseline levels specification. The dependent variable is the weekly change in product price per gram, and the regressors are the contemporaneous and lagged weekly changes in the commodity price per gram. The sample is a weekly ASIN panel of products from January 2021 to March 2025. Standard errors are two-way cluster-robust. T-statistics are computed using these standard errors.

Figure A.11: Coefficients, stacked specification: Cocoa

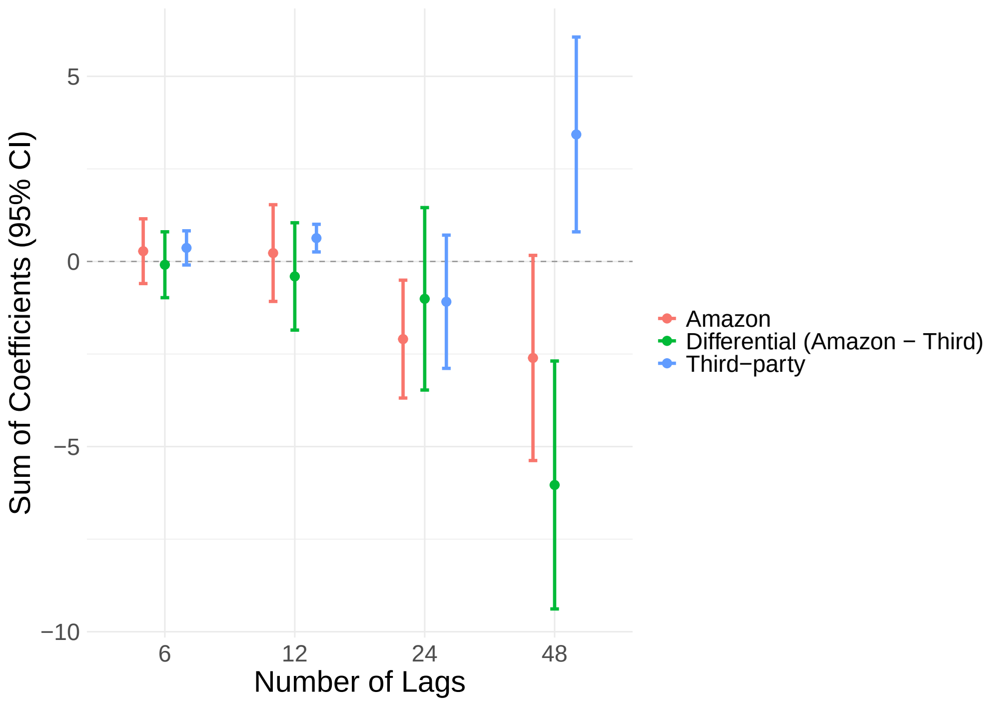


Table A.22: Pass-through: distributed lag regressions using Buy Box prices

Specification	K (lags)	Sum of Coefficients	Standard Error	T-statistic	Observations
Coffee:					
Buy Box 3P	6	-0.18	0.32	-0.56	1279218
Buy Box 3P	12	-0.37	0.49	-0.76	1279218
Buy Box 3P	24	-0.49	0.66	-0.75	1279218
Buy Box 3P	48	-0.18	0.66	-0.27	1279218
Buy Box Amazon	6	0.00	0.13	-0.04	275094
Buy Box Amazon	12	-0.15	0.18	-0.85	275094
Buy Box Amazon	24	-0.17	0.29	-0.61	275094
Buy Box Amazon	48	0.54	0.39	1.37	275094
Cocoa:					
Buy Box 3P	6	0.13	0.05	2.42	128352
Buy Box 3P	12	0.23	0.08	3.05	128352
Buy Box 3P	24	0.18	0.13	1.39	128352
Buy Box 3P	48	0.61	0.25	2.41	128352
Buy Box Amazon	6	0.03	0.10	0.26	9725
Buy Box Amazon	12	0.05	0.14	0.37	9725
Buy Box Amazon	24	-0.54	0.10	-5.27	9725
Buy Box Amazon	48	-1.08	0.38	-2.82	9725

Notes: Each row reports the cumulative pass-through estimate, $\sum_{k=0}^{K} \beta_k$, from the distributed-lag specification in equation 5, where K denotes the number of weekly lags of the commodity-price change (coffee and cocoa, respectively) included in the regression. Regressions are estimated separately for 3P seller prices conditionally on being shown in the Buy Box (*Buy Box 3P*) and Amazon prices conditionally on being shown in the Buy Box (*Amazon*) using the baseline levels specification. The dependent variable is the weekly change in product price per gram, and the regressors are the contemporaneous and lagged weekly changes in the commodity price per gram. The sample is a weekly ASIN panel of products from January 2021 to March 2025. Standard errors are two-way cluster-robust. T-statistics are computed using these standard errors.

Table A.23: Log pass-through: distributed lag regressions

Specification	K (lags)	Sum of Coefficients	Standard Error	T-statistic	Observations
Coffee:					
Third	6	0.02	0.01	2.16	1559229
Third	12	0.02	0.01	2.65	1559229
Third	24	0.04	0.01	3.31	1559229
Third	48	0.07	0.02	4.78	1559229
Amazon	6	-0.01	0.02	-0.65	288417
Amazon	12	-0.03	0.03	-1.09	288417
Amazon	24	-0.03	0.04	-0.81	288417
Amazon	48	0.02	0.04	0.54	288417
Cocoa:					
Third	6	0.01	0.01	1.47	149569
Third	12	0.03	0.01	2.70	149569
Third	24	0.05	0.02	2.87	149569
Third	48	0.11	0.02	4.70	149569
Amazon	6	0.02	0.02	0.73	11003
Amazon	12	0.02	0.04	0.49	11003
Amazon	24	-0.07	0.04	-1.86	11003
Amazon	48	-0.14	0.05	-3.09	11003

Notes: Each row reports the cumulative pass-through estimate, $\sum_{k=0}^{K} \beta_k$, from the distributed-lag specification in equation 5, where K denotes the number of weekly lags of the commodity-price change (coffee and cocoa, respectively) included in the regression. Regressions are estimated separately for logged 3P seller prices(*Third*) and logged Amazon prices (*Amazon*) using the baseline specification but in logs. The dependent variable is the weekly change in product price per gram, and the regressors are the contemporaneous and lagged weekly changes in the commodity price per gram. The sample is a weekly ASIN panel of products from January 2021 to March 2025. Standard errors are two-way cluster-robust. T-statistics are computed using these standard errors.

Table A.24: Pass-through: distributed lag regressions, ASINs operating under dual mode only

Specification	K (lags)	Sum of Coefficients	Standard Error	T-statistic	Observations
Coffee:					
Third	6	0.33	0.17	1.95	237995
Third	12	0.55	0.23	2.37	237995
Third	24	0.69	0.28	2.44	237995
Third	48	0.92	0.56	1.65	237995
Amazon	6	-0.05	0.14	-0.34	238252
Amazon	12	-0.12	0.18	-0.66	238252
Amazon	24	-0.25	0.26	-0.95	238252
Amazon	48	0.25	0.36	0.70	238252
Cocoa:					
Third	6	0.46	0.19	2.38	10676
Third	12	0.37	0.21	1.77	10676
Third	24	-0.06	0.41	-0.13	10676
Third	48	1.49	0.74	2.02	10676
Amazon	6	0.28	0.36	0.78	7467
Amazon	12	0.35	0.58	0.61	7467
Amazon	24	-2.17	0.62	-3.52	7467
Amazon	48	-2.01	1.06	-1.89	7467

Notes: Each row reports the cumulative pass-through estimate, $\sum_{k=0}^{K} \beta_k$, from the distributed-lag specification in equation 5, where K denotes the number of weekly lags of the commodity-price change (coffee and cocoa, respectively) included in the regression. Regressions are estimated separately for 3P seller prices(*Third*) and Amazon prices (*Amazon*) using the baseline levels specification. The dependent variable is the weekly change in product price per gram, and the regressors are the contemporaneous and lagged weekly changes in the commodity price per gram. The sample is a weekly ASIN panel of products from January 2021 to March 2025; this time, we only include ASINs for which in at least 10% of ASIN-weeks with any offer, Amazon and 3P sellers are jointly offering the product. Standard errors are two-way cluster-robust. T-statistics are computed using these standard errors.

Table A.25: Pass-through: distributed lag regressions, ASINs with at least 36 weeks of continuous 3P, or Amazon retail presence

Seller Type	K (lags)	Sum of Coefficients	Standard Error	T-statistic	Observations
Coffee:					
Third	6	0.32	0.21	1.55	185110
Third	12	0.59	0.28	2.11	185110
Third	24	0.70	0.36	1.94	185110
Third	48	1.15	0.67	1.72	185110
Amazon	6	0.00	0.16	-0.02	267065
Amazon	12	-0.16	0.20	-0.83	267065
Amazon	24	-0.17	0.30	-0.58	267065
Amazon	48	0.50	0.40	1.28	267065
Cocoa:					
Third	6	0.21	0.08	2.59	138821
Third	12	0.37	0.09	3.90	138821
Third	24	0.49	0.18	2.66	138821
Third	48	1.04	0.19	5.48	138821
Amazon	6	0.36	0.39	0.94	6896
Amazon	12	0.38	0.65	0.60	6896
Amazon	24	-2.12	0.73	-2.89	6896
Amazon	48	-1.73	1.48	-1.17	6896

Notes: Each row reports the cumulative pass-through estimate, $\sum_{k=0}^{K} \beta_k$, from the distributed-lag specification in equation 5, where K denotes the number of weekly lags of the commodity-price change (coffee and cocoa, respectively) included in the regression. Regressions are estimated separately for 3P seller prices(*Third*) and Amazon prices (*Amazon*) using the baseline levels specification. The dependent variable is the weekly change in product price per gram, and the regressors are the contemporaneous and lagged weekly changes in the commodity price per gram. The sample is a weekly ASIN panel of products from January 2021 to March 2025. In the 3P specifications, we only include ASINs that are observed for at least 36 consecutive weeks; likewise for the Amazon specifications. Standard errors are two-way cluster-robust. T-statistics are computed using these standard errors.